A Diagnostic Study of Gaussian Visual Robustness in JEPA Latent World Models

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Abstract

Latent predictive world models such as JEPAs predict future representations rather than pixels, but this objective alone does not specify visual robustness for closed-loop control. We study this gap as a diagnostic problem through action-conditioned predictive consistency (ACPC): corrupted and clean views of the same state may encode differently, but under the same action sequence their task-relevant predicted futures should agree while action-relevant state distinctions remain separable. In a controlled Gaussian-noise study on PushT, TwoRoom, Reacher, and Cube, we evaluate 36 epoch-10 LeWorldModel checkpoints with three evaluation seeds per cell and replicate the main signatures with PLDM. No-noise LeWM drops sharply under observation-only noise, for example from 86.33% to 4.33% on PushT and from 58.67% to 18.33% on Reacher. Full-sequence input-side noise augmentation recovers much of this loss, but the best recovery point and plateau structure vary by task. Post-hoc paired ACPC-basin probes localise the recovery: direct-evaluation-selected robust endpoints have smaller same-action prediction radii, e.g. PushT R_F decreases from 1.543 to 0.088. The paper contributes a diagnostic framing, release artifacts, and negative ablations for this controlled setting.

Keywords: visual world models; JEPA; action-conditioned predictive consistency; visual robustness; predictive-dynamics diagnostics.

1 Introduction

1.1 Latent prediction is not a robustness definition

JEPAs predict future representations instead of reconstructing pixels, which can reduce pressure to encode high-magnitude irrelevant visual detail [2, 4, 35]. For control, however, "irrelevant" is action- and transition-dependent. A representation may ignore a background texture but must preserve a contact pose or doorway state if that changes the next useful action. Thus encoder-level invariance is an incomplete target: planning uses the model's *predictions*, so robustness must be stated after action-conditioned prediction.

We use this failure as a controlled diagnostic probe. A no-noise LeWM [23] checkpoint scores 86.33% on unperturbed PushT but only 4.33% under observation-only Gaussian noise at std 0.08 with the goal unperturbed; Reacher drops from 58.67% to 18.33%, and TwoRoom from 94.00% to 65.67%. Observation+goal corruption is reported as a harder auxiliary stress condition. Visual-corruption fragility is not new [40, 39, 15, 24]; our focus is what it reveals about JEPA latent world-model control.

1.2 From visual invariance to action-conditioned predictive consistency

We reframe visual robustness for world-model control at the level of *action-conditioned predictive dynamics*. Let $z = E(h)$ and $\tilde{z} = E(\tilde{h})$ be the latents an encoder E produces from an unperturbed history h and a corrupted history $\tilde{h}$ of the same underlying state, and let $F^k(\cdot, \mathbf{a}_{t:t+k-1})$ be the action-conditioned latent predictor rolled out for k steps under an action sequence $\mathbf{a}$. We *allow* $z \neq \tilde{z}$. The robustness target is placed *after* the predictor: in task-relevant coordinates,

$$F^k\big(z, \mathbf{a}_{t:t+k-1}\big) \;\approx\; F^k\big(\tilde{z}, \mathbf{a}_{t:t+k-1}\big), \qquad k = 1, \ldots, H, \tag{1}$$

so that the two branches induce the same next-state and rollout predictions under the same action intervention. The requirement is *selective*: it should hold only for nuisance perturbations of the same underlying state. State differences that change action-conditioned transitions, costs, or the optimal action must remain *discriminable*, or the planner loses the distinctions it needs. We make both halves precise in Section 3.

This reframing changes what counts as a robustness measurement. Encoder-level latent closeness, $\|z - \tilde{z}\| \to 0$, is neither necessary (a large but task-irrelevant encoder shift can wash out after the predictor) nor sufficient (a small encoder shift can still flip the planned action). Planning, cost, and action metrics remain useful, but as *downstream readouts* of predictive consistency rather than as the central definition; future objectives should regularise the action-conditioned predictive dynamics.

1.3 Existing corruption results as a controlled probe

A natural remedy is input-side noise augmentation. We sweep $\sigma_{\max} \in \{0.01, \ldots, 0.08\}$ across four tasks and find recovery, but the useful range varies: TwoRoom reaches its highest robust values at high noise, PushT's unperturbed and noisy-eval point-best checkpoints differ, Reacher has a broad plateau, and Cube responds weakly. A single Gaussian-noise strength therefore recovers performance unevenly across tasks. We use this sweep to motivate future predictive-consistency objectives, not to identify a universal training-noise level. Appendix L records a negative target-view ablation: perturbed-history → original-future denoising is not a sufficient closed-loop fix, so the retained empirical branch is full-sequence perturbed-target training.

1.4 Contributions

C1 — Diagnostic principle and planner link. We formulate visual robustness for world-model control as action-conditioned predictive consistency with an action-relevant discriminability countercondition. Under a Lipschitz cost readout, bounded ACPC bounds candidate-cost drift on a fixed candidate set; with a sufficiently large clean action margin, it preserves the selected candidate within that set.

C2 — Controlled evidence. Within a fixed Gaussian-noise protocol, we find severe no-noise LeWM corruption cliffs and task-dependent recovery under input-side noise augmentation. PLDM replication checks whether the same signatures appear in a second model family.

C3 — Release package and negative checks. We release paired ACPC-basin artifacts, the full LeWM basin grid, cross-checkpoint diagnostic tables, PLDM replication, a target-view ablation, and a heteroscedastic-loss failure case. These artifacts localise what changed at selected checkpoints and expose where stronger readings fail.

2 Related Work

2.1 Latent prediction and JEPA world models

JEPA [21] predicts in latent space rather than reconstructing pixels; I-JEPA [2] and V-JEPA [4, 3] extend this idea to images and video. LeWM [23], our baseline, stabilises an end-to-end JEPA world model with SIGReg [7]. PLDM [29, 30], via `stable-worldmodel` [22], supplies the second method family in Appendix I. LeJEPA theory [18] studies latent-variable recovery and latent planning; our diagnostic asks whether learned visual world-model checkpoints give paired clean/corrupted views consistent action-conditioned predictions.

2.2 Invariance, augmentation, and visual robustness

Augmentation and joint-embedding losses impose invariances only when their training signal identifies the right nuisance factors [37, 32, 42]. Van Assel et al. [35] show that joint-embedding prediction can reduce pressure to encode high-magnitude irrelevant features, but still needs augmentation or bias to decide which variations are irrelevant. In control, that decision is action- and transition-conditioned. Seq-JEPA [11] addresses a related invariance–equivariance tension architecturally. DrQ/DrQ-v2 [40, 39], SODA/DMC-GB [15], DreamerV3 [13], TD-MPC2 [14], ViGMO [24], N-JEPA [25], variational VJEPA [16], and US-JEPA [26] provide visual-robustness and noisy-environment context; ACPC places the consistency requirement after action-conditioned rollout and pairs it with a discriminability guard.

2.3 Control-relevant predictive representations

Bisimulation merges states with equivalent reward and transition behaviour. DeepMDP [10], DBC [41], bisimulation-regularized MPC [28], and Bisim-JEPA [34] use this principle to learn control-relevant representations. Value-equivalent and value-aware models [12, 36] and VIBR [6] likewise emphasise downstream control consequences. Wang et al. [38] formalise action-conditioned video world modeling as a group action with identity, inverse, and composition-consistency metrics. These lines motivate our downstream-consequence view: same-state corrupted and unperturbed views should agree after prediction, while action-distinct transitions remain separable.

2.4 Representation diagnostics and collapse analysis

We use standard representation diagnostics: effective rank [27, 17, 33], CKA [19], linear probes [1], and anti-collapse context from VICReg [5]. RankMe [9] motivates our cross-checkpoint question: after removing the sweep-level training-noise trend, does a label-free predictor-sensitivity diagnostic retain downstream control signal? The individual metrics are standard; our contribution is the protocol that ties them to closed-loop Gaussian-corruption evaluation. A compact boundary table is provided in Appendix A.

3 Action-Conditioned Predictive Consistency

This section defines action-conditioned predictive consistency and its discriminability countercondition. Section 3.2 then separates the main descriptive diagnostics from paired ACPC probes.

3.1 Setup and definitions

Let o_t be an unperturbed observation and $\tilde{o}_t = \tau(o_t)$ a visually corrupted view of the *same* underlying state (τ is additive Gaussian pixel noise here, but the definitions are corruption-agnostic). Let h_t and $\tilde{h}_t$ be the corresponding observation–action histories, E_θ the encoder, and F_θ the action-conditioned latent predictor. Fix an action sequence $\mathbf{a}_t^H = a_{t:t+H-1}$ that is *identical* on both branches. For compact notation below, $\mathbf{a}_{0:k-1}$ denotes the length-k prefix of this sequence after re-indexing time at t, and we write

$$z_t = E_\theta(h_t), \quad \tilde{z}_t = E_\theta(\tilde{h}_t), \qquad \hat{z}_{t+k} = F_\theta^k(z_t, \mathbf{a}_{0:k-1}), \quad \hat{\tilde{z}}_{t+k} = F_\theta^k(\tilde{z}_t, \mathbf{a}_{0:k-1}). \tag{2}$$

Let Π denote a *task-relevant predictive readout* — the full latent, a transition delta, goal-distance or cost features, or a learned projection — and let d be a distance on its range.

Action-conditioned predictive consistency. For a same-state perturbation pair under the shared action sequence,

$$\mathrm{ACPC}_H(o_t, \tilde{o}_t, \mathbf{a}) = \sum_{k=1}^{H} \alpha_k \, d\big(\Pi(\hat{z}_{t+k}), \, \Pi(\hat{\tilde{z}}_{t+k})\big), \qquad \alpha_k \geq 0. \tag{3}$$

In this diagnostic framing, robustness corresponds to ACPC_H being small, not to the encoder distance $d(z_t, \tilde{z}_t)$ being small. Encoder closeness is neither necessary (a large but task-irrelevant encoder shift can wash out through F_θ and Π) nor sufficient (a small encoder shift can still change the predicted future enough to flip the planned action).

Action-relevant discriminability (countercondition). Consistency must be *selective*. Let $Y^H(s, \mathbf{a})$ denote a task-relevant future readout that is external to the learned predictor — simulator state, cost-to-go, inverse-dynamics label, contact/keyframe tag, or another diagnostic proxy available in the dataset. For two genuinely different states s_i, s_j (with latents z_i, z_j) that admit an action sequence under which this external future readout diverges,

$$\Delta_{\mathrm{dyn}}^H(s_i, s_j, \mathbf{a}) = d_Y\big(Y^H(s_i, \mathbf{a}), \, Y^H(s_j, \mathbf{a})\big) > m, \tag{4}$$

the representation and predictions should keep them separable,

$$d\big(\Psi(z_i), \, \Psi(z_j)\big) > m', \tag{5}$$

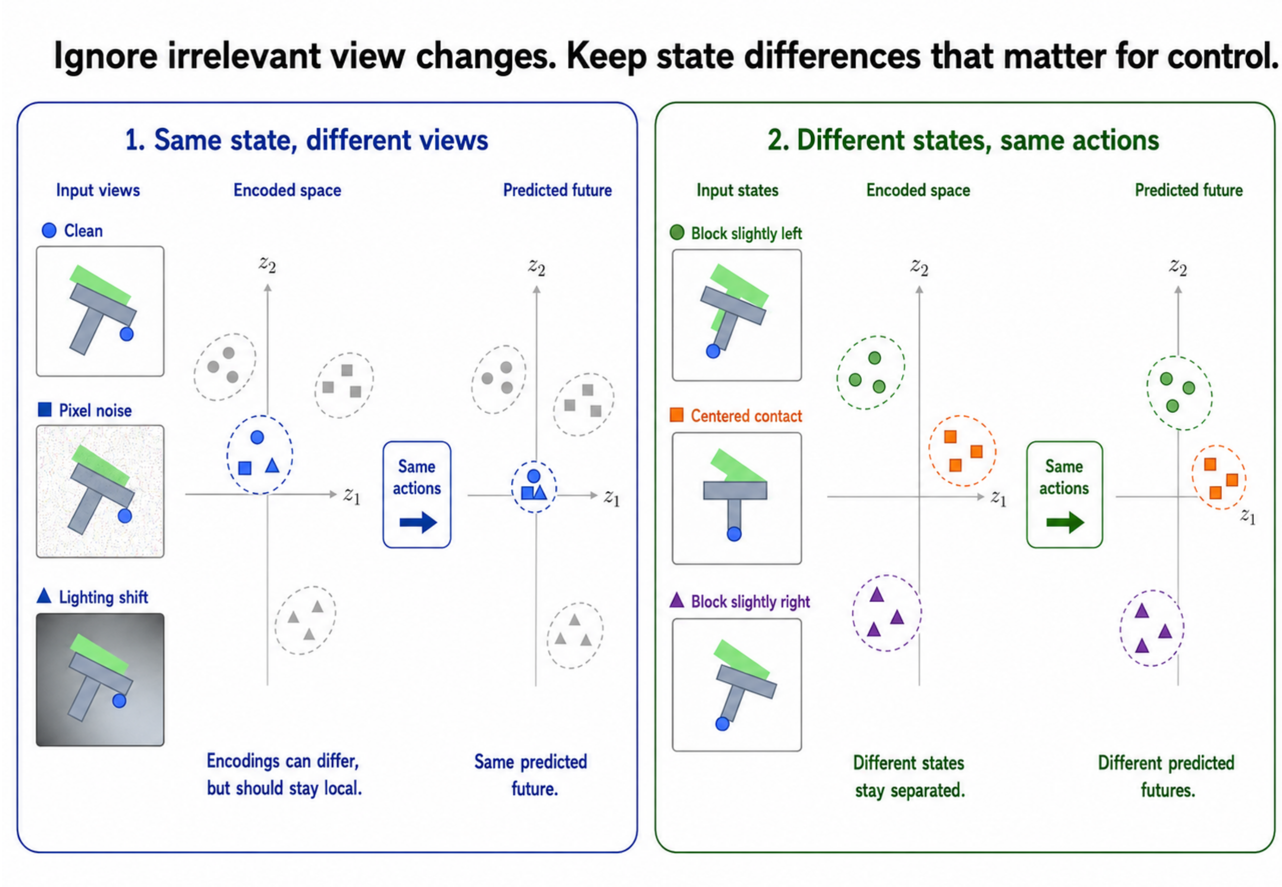

Figure 1: Conceptual reading of the selective-consistency tension. Same-state nuisance variants may encode differently but should induce consistent action-conditioned predictions under the same action sequence, while transition-, cost-, or action-relevant state differences must remain resolvable.

where Ψ is a discriminability readout (latent, predicted delta, inverse-dynamics feature, cost feature, or rollout embedding) and $m, m' > 0$ are margins. Equations (3)–(5) state the target: lower predictive disagreement for same-state visual perturbations must preserve discriminability among state/action/transition-distinct pairs. A method that drove $\mathrm{ACPC}_H \to 0$ by collapsing the latent would violate (5), so the two conditions are inseparable.

3.2 Operational consistency diagnostics

The main paper uses one fixed ACPC instance. For the Gaussian-noise basin diagnostic, Π is the identity readout on the model inference/cost latent rollout, d is Euclidean L2 over rollout tokens, and $H = 8$. The reported quantities are R_E, the median same-state encoder-history spread normalised by the original-view nearest-neighbour L2 scale, and R_F, the median same-state rollout-latent spread normalised by the unperturbed transition L2 reference over the same horizon. R_F/R_E is a descriptive contraction ratio.

The main text also keeps three non-ACPC measurements: closed-loop success, the encoder-shift ratio R_E, and proxy discriminability measurements such as effective rank, transition-resolution ratio, and inverse-dynamics probe R^2. Predictive cost consistency, candidate-ranking agreement, margin-conditioned action flips, ADM, and SPRR are exploratory downstream readouts defined in Appendix K.1; they are not used for model selection or for the main robustness claim.

3.3 ACPC as a fixed-candidate stability condition

ACPC is connected to planning through the candidate costs evaluated by model-predictive control. Let h be a clean history, $\tilde{h}$ the corresponding corrupted history, and let both branches evaluate the same candidate action sequence $\mathbf{a}$. For a goal or task descriptor g, define

$$C_h(\mathbf{a}, g) = J(\Pi(\hat{z}_{t+1}), \dots, \Pi(\hat{z}_{t+H}), g)\,, \tag{6}$$

where J is the planner's cost readout on the rollout-readout sequence. $C_{\tilde{h}}(\mathbf{a}, g)$ is defined analogously from the corrupted branch. The following statements are sufficient conditions for stability on a fixed candidate set; they do not prove stability of the full CEM sampling distribution or of the closed-loop trajectory over repeated replanning.

Proposition 1 (ACPC controls candidate-cost drift)**.** *Assume that J is L_J-Lipschitz in the rollout-readout sequence under metric d_H. If*

$$d_H\big(\Pi(F_\theta^{1:H}(E_\theta(h), \mathbf{a})), \Pi(F_\theta^{1:H}(E_\theta(\tilde{h}), \mathbf{a}))\big) \le \epsilon,$$

then

$$|C_h(\mathbf{a}, g) - C_{\tilde{h}}(\mathbf{a}, g)| \le L_J \epsilon.$$

Proof. The inequality is the Lipschitz condition applied directly to the clean and corrupted rollout-readout sequences. □

In the reported basin diagnostic, this condition is instantiated with the L2 rollout-token distance underlying R_F; other downstream readouts are treated only as exploratory diagnostics.

Proposition 2 (Candidate top-1 stability)**.** *Let $\mathcal{A} = \{\mathbf{a}^1, \dots, \mathbf{a}^K\}$ be the shared candidate set. Assume lower cost is preferred, and let $\mathbf{a}^{(1)}$ and $\mathbf{a}^{(2)}$ denote the lowest- and second-lowest-cost candidates on the clean branch. Suppose that for every candidate j,*

$$|C_h(\mathbf{a}^j, g) - C_{\tilde{h}}(\mathbf{a}^j, g)| \le \eta.$$

If the clean branch has a strict top-1/top-2 margin

$$\Delta = C_h(\mathbf{a}^{(2)}, g) - C_h(\mathbf{a}^{(1)}, g) > 2\eta,$$

then the corrupted branch selects the same top-1 candidate.

Proof. For any candidate $\mathbf{a}^j \ne \mathbf{a}^{(1)}$,

$$C_{\tilde{h}}(\mathbf{a}^j, g) - C_{\tilde{h}}(\mathbf{a}^{(1)}, g) \ge C_h(\mathbf{a}^j, g) - C_h(\mathbf{a}^{(1)}, g) - 2\eta \ge \Delta - 2\eta > 0.$$

Thus every non-top candidate remains more costly than the clean top-1 candidate on the corrupted branch. □

Corollary 1 (ACPC-margin condition for candidate stability)**.** *Let $\mathcal{A} = \{\mathbf{a}^1, \dots, \mathbf{a}^K\}$ be a shared candidate set and assume the cost readout J is L_J-Lipschitz as in Proposition 1. If every candidate in $\mathcal{A}$ has rollout-readout discrepancy at most ϵ between the clean and corrupted branches, and the clean branch has top-1/top-2 margin $\Delta > 2L_J\epsilon$, then the two branches select the same top-1 candidate from $\mathcal{A}$.*

Proof. Proposition 1 gives $|C_h(\mathbf{a}^j, g) - C_{\tilde{h}}(\mathbf{a}^j, g)| \le L_J\epsilon$ for every candidate. Applying Proposition 2 with $\eta = L_J\epsilon$ gives the result. □

The corollary gives a formal link between ACPC and planning without turning the diagnostic into a closed-loop guarantee. It connects paired predictive agreement to candidate-cost stability and then to action selection under a margin condition. Candidate-ranking agreement and margin-conditioned action flips are downstream readouts of this chain. The result is limited to a fixed candidate set; CEM resampling, repeated replanning, and environment feedback can still change the closed-loop trajectory.

Encoder closeness is not the right substitute for this condition. It is not necessary: a nuisance-subspace change can move $E_\theta(h)$ and $E_\theta(\tilde{h})$ far apart while leaving $\Pi(F_\theta^{1:H}(\cdot, \mathbf{a}))$ and the cost unchanged. It is not sufficient either: a small encoder displacement can flip a low-margin cost ranking when the predictor or cost readout has high local sensitivity.

Proposition 3 (ACPC alone permits collapse)**.** *There exist encoder–predictor pairs for which* $\mathrm{ACPC}_H = 0$ *for every same-state perturbation pair and every action sequence, while action-distinct states are mapped to identical rollout readouts.*

Proof. Let $E_\theta(h) = z_0$ for every history and let $F_\theta^k(z_0, \mathbf{a}) = u_0$ for every horizon k and action sequence $\mathbf{a}$. Then clean and corrupted branches always have identical rollout readouts, so same-state ACPC is zero. However, any two states that require different actions, costs, or transitions also have identical readouts, violating the discriminability condition in Equation (5). □

Thus low ACPC is meaningful only with a guard on action-relevant separation. This is the role of the transition-resolution and inverse-dynamics probes in the main diagnostics, and it is the failure exposed by the heteroscedastic-loss ablation in Appendix G.

4 Study protocol

4.1 Models and noise intervention

LeWM [23] is an end-to-end JEPA world model trained with a latent prediction loss and SIGReg regularisation; inference uses CEM for latent-space MPC. PushT (2D pushing), TwoRoom (2D navigation), Reacher (2D arm), and Cube (3D manipulation) form a heterogeneous stress panel spanning different nuisance/discriminability regimes. We add per-frame Gaussian noise to the input sequence as a full-sequence augmentation: each frame is independently noised with probability $p = 1.0$, and the standard deviation is drawn from Uniform$(0, \sigma_{\max})$. Because the target future frame is also encoded from the noised sequence, this intervention is not an original-target denoising objective. We sweep $\sigma_{\max} \in \{0.01, \ldots, 0.08\}$ and evaluate observation-only Gaussian noise with the goal image unperturbed as the primary robustness endpoint. Observation+goal corruption is retained as an auxiliary stress condition.

4.2 Evaluation and diagnostics

The canonical LeWM grid contains 36 epoch-10 checkpoints: four tasks and, for each task, a no-noise baseline plus eight noise-trained checkpoints. Each checkpoint is evaluated with three evaluation seeds (42/43/44) and 100 trajectories per seed; these are not independent training seeds. Tables report across-seed population mean ± std from the canonical evaluation artifact. The runs use a single NVIDIA H800 (80 GB). The main text uses four result blocks: no-noise corruption cliffs, the noise-training sweep, paired ACPC-basin radii, and cross-checkpoint diagnostic limitations. The longer five-layer diagnostic definitions, latent-noise mechanism probes, exact implementation keys, and full tables are in Appendix C and Appendix B.10.

For cross-checkpoint checks we compute Spearman ρ across the 9 LeWM checkpoints in each task and partial Spearman conditioned on $\sigma_{\max}$. The partial step removes the sweep-level training-noise trend, asking whether a diagnostic retains residual checkpoint-quality signal. We report 95% percentile bootstrap intervals ($B = 1000$, seed $= 42$) in the released artifact and use the main table only as a small-sample diagnostic check. PLDM replication is reported in Appendix I.

5 Experiments

All results use the protocol in Section 4: LeWM-base denotes the no-noise checkpoint, and LeWM+noise denotes the eight-level $\sigma_{\max}$ sweep. We first report closed-loop corruption behaviour, then the paired ACPC-basin diagnostic, and finally cross-checkpoint diagnostic limitations.

5.1 JEPA control fragility under visual corruptions

Table 1 reports LeWM-base success rates under unperturbed and noised evaluation (mean ± std across 3 seeds × 100 evaluations).

LeWM-base is strong on unperturbed images (especially TwoRoom and PushT), but observation noise alone already produces large closed-loop drops: PushT loses 82.00 pts at observation-noise 0.08, Reacher 40.33 pts, TwoRoom 28.33 pts, and Cube 19.67 pts. The observation+goal column shows the stronger full-visual-stream stress case; it further degrades TwoRoom and Reacher, but is not the primary robustness endpoint because the goal image is also perturbed. The drop pattern across tasks remains informative: Cube degrades least, while PushT degrades most sharply, consistent with contact-heavy continuous control being most sensitive to visual precision.

Table 1: LeWM-base under test-time visual corruptions (mean ± std; 3 seeds × 100 evaluations).

Task	unpert.	obs 0.05	obs 0.08	obs+goal 0.08	drop
TwoRoom	94.00 ± 3.56	72.33 ± 5.79	65.67 ± 1.25	50.00 ± 1.41	-28.33
PushT	86.33 ± 2.36	17.00 ± 1.41	4.33 ± 1.25	4.67 ± 2.05	-82.00
Reacher	58.67 ± 1.25	33.67 ± 2.87	18.33 ± 2.05	15.00 ± 2.16	-40.33
Cube	66.67 ± 2.62	48.67 ± 6.85	47.00 ± 6.38	46.33 ± 3.68	-19.67

Table 2: LeWM noise-sweep summary. Values are success rate mean ± population std across evaluation seeds; parenthesised values are the train-time $\sigma_{\max}$ of the point-best checkpoint.

Task	unpert. base	obs 0.08 base	unpert. best	obs 0.08 best	Main reading
TwoRoom	94.00 ± 3.56	65.67 ± 1.25	98.33 ± 0.47 (0.08)	97.67 ± 0.94 (0.08)	high-noise plateau
PushT	86.33 ± 2.36	4.33 ± 1.25	89.67 ± 1.70 (0.03)	89.00 ± 4.32 (0.08)	unperturbed and noisy point-bests differ
Reacher	58.67 ± 1.25	18.33 ± 2.05	86.00 ± 2.94 (0.06)	84.67 ± 5.19 (0.07)	broad robust plateau
Cube	66.67 ± 2.62	47.00 ± 6.38	69.33 ± 0.47 (0.01)	68.33 ± 2.49 (0.03)	weak and non-monotone response

PLDM provides a second-family boundary on this failure mode. Without noise augmentation, PLDM drops sharply on PushT ($75.33\% \pm 3.68$ unperturbed to $18.33\% \pm 2.05$ at observation-noise 0.08, -57.00 pt) and TwoRoom ($96.67\% \pm 1.70$ to $67.00\% \pm 2.16$, -29.67 pt), while Reacher and Cube have much smaller no-noise-training gaps (-2.33 and -4.33 pt; Table 17). The noise-training signatures still carry over on the tasks with substantial cliffs: TwoRoom recovers into a high-noise plateau and PushT has a large $\sigma_{\max} \approx 0.03$ recovery; Cube remains weakly responsive (Section I).

5.2 Noise augmentation partly closes the gap with task-dependent plateaus

Table 2 summarises the sweep points needed for the main argument; the complete 8-level tables are in Appendix B.4. All entries use the unified $n = 3$ seeds (42/43/44) × 100 trajectories protocol.

The sweep has two readings. First, input-side noise can be effective in this protocol: PushT recovers from 4.33% to 89.00% at observation-noise 0.08, and Reacher from 18.33% to 84.67%. Second, the useful augmentation strength is task-dependent. TwoRoom favours heavy noise, PushT's unperturbed and noisy-eval point-bests differ (0.03 vs 0.08), Reacher has a broad plateau, and Cube responds weakly. Several point-bests sit within plateau regions rather than isolated peaks; the robust pattern is that one scalar perturbation strength expresses the selective-consistency tradeoff only coarsely.

The behavioural evidence for this selective-consistency tension is summarised by Table 2 and fig. 2. Before turning to the broader five-layer diagnostic suite, we report a paired Gaussian-noise ACPC-basin proxy that tests whether same-state noisy views occupy a smaller action-conditioned predictive region after noise training.

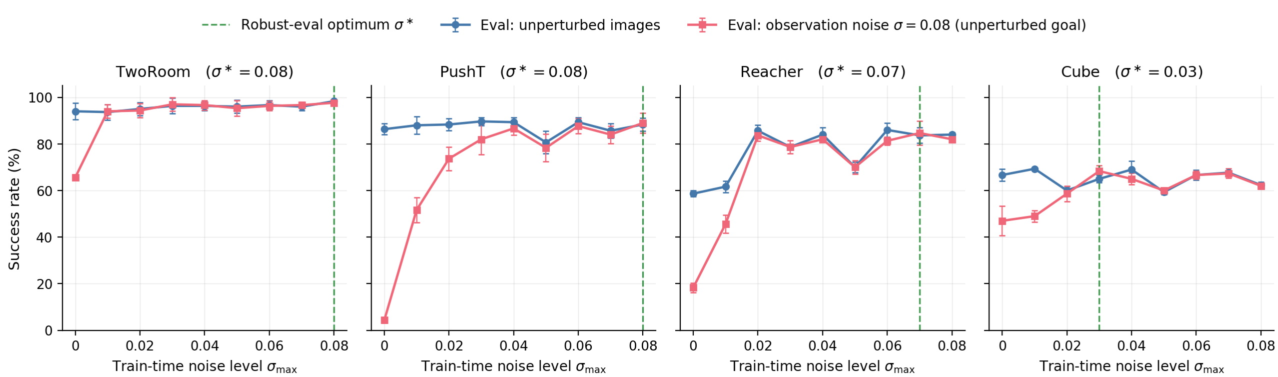

Figure 2: Noise-training sweep across four tasks. Blue: unperturbed evaluation; red: observation-only Gaussian noise $\sigma = 0.08$ with an unperturbed goal; green dashed: each task's corrupted-eval point-best $\sigma_{\max}$. The useful noise range varies by task, so a single scalar perturbation strength does not encode the nuisance/action distinction.

Table 3: LeWM Gaussian-noise ACPC-basin proxy: no-noise baseline vs. each task's observation-noise 0.08 point-best checkpoint. R_E is normalised encoder-view spread and R_F is normalised action-conditioned rollout-view spread under the same action sequence.

Task	checkpoint	obs 0.08	drop	R_E	R_F
TwoRoom	base	65.67	28.33	3.538	0.785
TwoRoom	noise 0.08	97.67	0.67	0.235	0.028
PushT	base	4.33	82.00	1.727	1.543
PushT	noise 0.08	89.00	−0.67	0.137	0.088
Reacher	base	18.33	40.33	4.238	1.012
Reacher	noise 0.07	84.67	−1.00	0.135	0.027
Cube	base	47.00	19.67	1.343	0.957
Cube	noise 0.03	68.33	−3.33	0.238	0.115

Table 4: Compact representation-diagnostic movement: LeWM-base → representative noise-trained checkpoint. Representative $\sigma_{\max}$ values are the original diagnostic checkpoints (TwoRoom 0.08, PushT 0.02, Reacher 0.06, Cube 0.01).

Task	rep. $\sigma_{\max}$	NN cos dist	eff. rank	trans. L2	rollout T8 L2
TwoRoom	0.08	0.0449 → 0.0321	47.60 → 37.69	0.7216 → 0.6621	18.62 → 0.66
PushT	0.02	0.2360 → 0.2477	76.42 → 77.41	0.3015 → 0.2989	18.65 → 6.00
Reacher	0.06	0.0633 → 0.0676	61.04 → 65.92	0.3704 → 0.3791	15.17 → 0.44
Cube	0.01	0.1856 → 0.1879	73.25 → 71.83	0.4847 → 0.4629	20.20 → 19.25

5.3 Paired Gaussian-noise ACPC basin diagnostic

We compute an eval-only ACPC basin diagnostic on the same 36 LeWM epoch-10 checkpoints used in the sweep. For each checkpoint we sample 100 fixed dataset windows and create eight paired same-state views using Gaussian pixel noise with std $\in \{0.01, \ldots, 0.08\}$, matching the training sweep's corruption family. Each original/noised branch is encoded and then rolled out under the same recorded action sequence for horizon $H = 8$. The reported prediction readout is the identity on the resulting inference/cost latent rollout sequence, with L2 distance over rollout tokens. The released source and rendering script are listed in Appendix B.10. The diagnostic intentionally excludes blur/resize so that the ACPC evidence is not confounded by train/eval corruption-family mismatch.

For each checkpoint, let R_E be the median pairwise spread among the original and noised encoder-history views, normalised by the original-view local NN L2 scale, and let R_F be the median pairwise spread among their $H = 8$ predicted rollout latents, normalised by the unperturbed transition L2 reference over the same horizon. Lower R_F means the same-state perturbation views induce a tighter action-conditioned predictive basin; R_F/R_E is a descriptive contraction ratio rather than a success predictor.

This comparison is diagnostic, not predictive: the robust endpoint is selected by direct closed-loop evaluation, and the basin radius is used afterward to localise what changed. Appendix K provides an exploratory downstream check under the same shared-candidate action protocol.

The direct-evaluation-selected endpoints have much smaller same-action prediction radii under same-family visual perturbations: PushT reduces R_F from 1.543 to 0.088, and TwoRoom from 0.785 to 0.028. Figure 3 shows the representative PushT contraction, and Appendix B.8 adds a projection-free local-atlas check of the same data. The comparison should be read as a post-hoc local diagnostic. It does not supply a model-selection rule, and it does not show that encoder representations remain separated while only predictions contract. In this Gaussian-noise setting, robust checkpoints usually shrink both R_E and R_F; the rollout radius is the quantity tied to the ACPC readout.

The broader representation-level evidence appears next in Table 4.

5.4 Diagnostic analysis: why one noise strength is only a coarse control

The per-task sweep shows where Gaussian noise helps. The diagnostics ask which latent properties move under noise training and whether basic action-relevant information survives. Table 4 compares core diagnostic metrics on LeWM-base versus the per-task representative noise-trained diagnostic checkpoint.

Notes. (i) All representative diagnostics are taken from the corresponding checkpoint's per-tool JSON outputs under the diagnostics directory (max-std = 0.1, history-only noise); (ii) the near-collapse of multi-step rollout

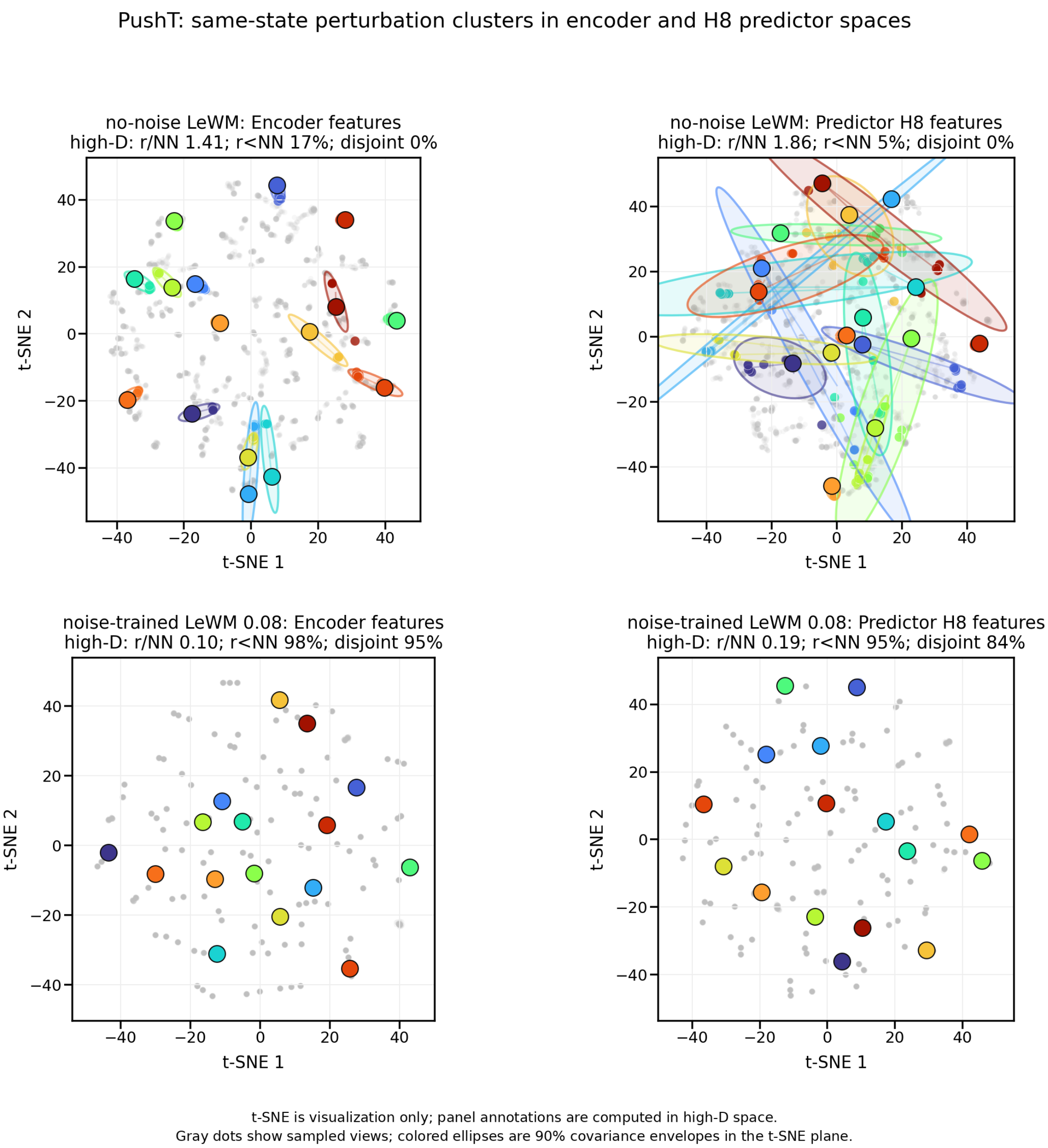

Figure 3: Representative PushT same-state perturbation clusters for Table 3. Top: no-noise-trained LeWM; bottom: noise-trained LeWM at $\sigma_{\max} = 0.08$. Left: encoder-history features; right: $H = 8$ predictor-rollout features. t-SNE coordinates and envelopes are visualization only; the small panel summaries report original-space local statistics.

Table 5: LeWM $n = 9$ sweep: per-task Pearson r / Spearman ρ vs corruption drop (unperturbed − observation-noise 0.08, unperturbed goal). Eval values come from the unified 3-seed × 100 protocol.

Metric	TwoRoom (r/ρ)	PushT (r/ρ)	Reacher (r/ρ)	Cube (r/ρ)
fragility ratio	+0.92/+0.20	−0.55/−0.80	+0.98/+0.55	+0.39/+0.27
rollout drift	+0.81/+0.31	+0.98/+0.93	+0.99/+0.58	+0.91/+0.48

drift at the heavier-noise representatives (TwoRoom 18.62 → 0.66 at $\sigma_{\max} = 0.08$; Reacher 15.17 → 0.44 at $\sigma_{\max} = 0.06$) is not a recording error: these checkpoints sharply reduce multi-step predictor drift while leaving the single-step task-resolution metrics comparatively flat — precisely the dissociation that motivates the analysis in Section 5.5. The low-noise representatives reduce drift far less (PushT 18.65 → 6.00 at $\sigma_{\max} = 0.02$; Cube 20.20 → 19.25 at $\sigma_{\max} = 0.01$).

Mechanistic reading.

- **TwoRoom.** Low-dimensional, discrete, visually redundant. The observed compression (effective rank 47.6 → 37.7) is compatible with high success in this task; the data do not show that compactness itself causes the improvement.
- **PushT.** Continuous contact requires fine-grained pose resolution. Noise training largely respects this bound: at the representative checkpoint ($\sigma_{\max} = 0.02$) the effective rank and inverse-dynamics probe stay flat (76.4 → 77.4; 0.774 → 0.769), and the L2 transition-resolution metric declines only mildly and monotonically toward heavier noise (0.3015 at base, 0.2989 at 0.02, 0.2789 at 0.08; full per-checkpoint diagnostics are released with the diagnostic artifacts). The catastrophic resolution erasure on PushT appears under error-based reweighting (Section G), not under input-side noise.
- **Reacher.** Low-dimensional continuous reaching does not show a resolution collapse in Table 4: effective rank, transition resolution, and the controllability probe remain flat or slightly improve. The notable change is the large reduction in multi-step predictor drift (15.17 → 0.44), matching the later finding that Reacher's residual corruption-drop signal lives in the rollout metric rather than in the single-step fragility ratio.
- **Cube.** Cube is moderately resolution-demanding but less fragile than PushT. The representative checkpoint leaves rank, transition resolution, controllability probe, and rollout drift almost unchanged, consistent with Cube's smaller visual-corruption cliff and the moderate residual signal of multi-step drift in Table 6.
- **Predictor-rollout caveat.** A drop in multi-step predictor rollout drift is not unambiguously good news. It can also mean the latent has become more *predictable* without being more *controllable*: predictor stability bought by sacrificing resolution.

These proxy discriminability checks are the main-text guard against a collapse-only reading. PushT is the sharpest case: the robust input-noise representative keeps effective rank and ID probe essentially flat, while the heteroscedastic-loss ablation in Appendix G shows the failure mode when error-based reweighting suppresses action-critical contact transitions. We therefore claim a proxy-level discriminability guard, not an oracle margin proof.

5.5 Cross-checkpoint correlation analysis

Table 5 reports task-specific cross-checkpoint correlations on the LeWM $n = 9$ sweep using the unified 3-seed × 100 evaluation.

We analyse two single-value-per-checkpoint diagnostic metrics that have full coverage on all 9 checkpoints per task: the fragility ratio, defined in Section C.2 as the single-step predictor target shift normalised by the nearest-neighbour cosine distance at the diagnostic's maximum injected std, and the corresponding multi-step rollout L2 drift. Both are pure checkpoint-level quantities derived from predictor-sensitivity diagnostics, independent of the evaluation protocol; exact artifact names are listed in Appendix B.10.

Reading. Unconditional correlations are large because both diagnostic metrics and the corruption drop are driven by the same upstream variable — $\sigma_{\max}$. The relevant test is partial correlation conditioned on $\sigma_{\max}$ (Table 6).

After removing the monotone sweep trend associated with $\sigma_{\max}$, the **fragility metric carries little stable information about the size of the corruption drop** on any task; the bootstrap intervals are wide and include, or round to, zero. The **multi-step predictor drift** retains only weak residual signal under the unperturbed-goal

Table 6: Partial Spearman ρ(metric, corruption drop | $\sigma_{\max}$), $n = 9$ per task.

Metric	TwoRoom	PushT	Reacher	Cube
fragility ratio	-0.03	$+0.19$	$+0.27$	$+0.12$
rollout drift	0.00	-0.01	$+0.37$	$+0.23$

observation-noise 0.08 endpoint (Reacher +0.37, Cube +0.23, both with wide intervals). This weakens the older observation+goal stress reading and is the reason we treat the cross-checkpoint diagnostic as a local explanation aid, not as a robustness predictor.

The PushT detailed table and scatter plot are moved to Appendix E. They show the same split at higher resolution: fragility ratio carries residual checkpoint-quality signal for unperturbed and noisy success after conditioning on $\sigma_{\max}$, but it does not isolate the corruption gap itself.

Appendix H reports the latent-noise mechanism probe. Its role is supporting, not central: the strongest common signal is encoder-side perturbation transduced by the predictor, but the present data do not isolate planner or cost-surface causality.

6 Discussion and limitations

Scope. This paper is a controlled diagnostic study. The canonical LeWM grid uses three evaluation seeds per checkpoint, not independent training seeds. Gaussian pixel noise is the main training and evaluation axis; blur is an eval-only sanity check in Appendix J. The ACPC basin probe is computed after closed-loop endpoint selection and is not a model-selection rule. The discriminability condition is tested through proxy diagnostics rather than oracle state margins. The main mechanism analysis is LeWM-centred, with PLDM used as a replication and scope check.

Interpretation. The data challenge a too-strong reading of latent prediction: predicting future latents does not by itself guarantee robustness to visual noise. The corrective is also not encoder invariance. The useful target is agreement of action-conditioned predictions in task-relevant coordinates while retaining distinctions that affect actions, costs, or transitions. This reading is compatible with the task variation: TwoRoom can tolerate substantial nuisance contraction, PushT contact control needs fine pose resolution, Reacher sits in a lower-dimensional continuous regime, and Cube is less sensitive to global Gaussian noise in this sweep.

Using the diagnostics. The diagnostics can help triage checkpoints within this protocol, but final model choice still requires closed-loop corruption evaluation. The PushT partial-correlation split is the main warning: after conditioning on $\sigma_{\max}$, fragility ratio still carries residual checkpoint-quality signal for unperturbed and noisy success, yet it does not isolate the corruption gap. The ACPC basin table has the same status. The compact base-vs-selected endpoint view localises what changed at already selected checkpoints, while the full grid in Appendix B.5 shows that the basin radius is not a standalone success predictor.

Negative checks and next steps. Two negative checks bound the method story. The heteroscedastic NLL probe collapses PushT because contact-control transitions are high-error and action-critical; prediction difficulty alone is the wrong gate. The target-view ablation in Appendix L rules out perturbed-history $\rightarrow$ original-future targets as a sufficient closed-loop fix. The natural next experiment is therefore a trained paired predictive-dynamics consistency objective after the action-conditioned predictor, gated by action sensitivity and protected by a discriminability guard. Broader replication should include frozen or pretrained visual features, EMA-target or variational JEPA variants, and non-Gaussian training corruptions.

7 Conclusion

This paper studies Gaussian visual robustness in JEPA latent world-model control through action-conditioned predictive consistency with an explicit discriminability countercondition. Controlled Gaussian corruptions show that latent prediction alone does not guarantee closed-loop robustness. Input-side noise augmentation recovers much of the lost control performance, but with task-dependent plateaus. The paired ACPC basin diagnostic adds a local mechanism read: direct-eval-selected robust endpoints have smaller same-action prediction radii under same-state perturbations, while pointwise fragility does not isolate the corruption gap after conditioning on train-time noise.

The target-view ablation in Appendix L rules out perturbed-history → original-future targets as a sufficient closed-loop fix and leaves full-sequence perturbed targets as the empirical branch for this study. The next method question is paired predictive-dynamics consistency after the action-conditioned predictor, gated by action sensitivity and protected by discriminability.

Acknowledgements

The complete code and data for this revision are available at `https://github.com/Anguo-star/le-wm/tree/ag/dev`; JSON artifact hashes are listed in the data manifest. We thank the LeWM authors for releasing the baseline framework on which this study builds.

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A Related-work boundary table

Table 7: Boundary to neighbouring consistency and control-representation work.

Neighbouring thread	Shared concern	Boundary in this paper
ViGMO [24]	Visual distractions and latent consistency	We do not propose a generic encoder-consistency regulariser; our diagnostic places consistency after action-conditioned rollout and adds a discriminability guard.
Bisimulation / Bisim-JEPA [10, 41, 34]	Control-relevant state abstraction	Same-state corrupted and unperturbed views need not share an encoder latent; agreement is required at the predictive-dynamics level.
LeJEPA theory [18]	Latent variables and latent planning	We do not claim identifiability; we test whether learned checkpoints preserve predictive consistency under visual perturbation.
Group-action world models [38]	Action-faithful rollout structure	We test same-action paired corruption consistency, not identity/inverse/composition consistency across action sequences.
Value-aware models [12, 36]	Models should serve downstream control	We make this operational as paired action-conditioned rollout agreement, not value equivalence or value prediction.
V-JEPA family [4, 3, 16]	Latent video prediction and noisy-environment probes	Representation recovery and closed-loop control stress different properties; our probe measures eval-time closed-loop behaviour.

B Experimental details

B.1 Environments

- **PushT.** 2D continuous pushing; 20,000 expert episodes; $\sim$196 steps/episode; action dim 2 (direction + magnitude); 224×224 RGB.
- **TwoRoom.** 2D continuous navigation; 10,000 episodes; $\sim$92 steps; action dim 2; 224×224 RGB.
- **Reacher.** 2D arm control (DeepMind Control Suite); 10,000 episodes; 200 steps; action dim 2.
- **Cube.** 3D cube manipulation (OGBench); 10,000 episodes; 200 steps; action dim 7.

B.2 Noise augmentation implementation

The actual implementation is `utils.py::AddNormalizedGaussianNoise`. Key points: (i) noise is added on ImageNet-normalized tensors and must be divided by the per-channel std to retain "pixel-space std" semantics; (ii) sampling is per-frame independent over the leading frame dims, not per-batch; (iii) all sweeps in this paper fix the corruption probability to 1.0, set the lower std to 0, and vary only the upper std $\sigma_{\max}$.

```
class AddNormalizedGaussianNoise:
    """Per-frame independent. Each frame draws Bernoulli(noise_prob) then
    std ~ Uniform(std_min, std_max). Pixel-space std is converted to
    normalized space by dividing by per-channel std before adding."""
    def __init__(self, std_min, std_max, noise_prob=1.0):
        self.std_low, self.std_high, self.noise_prob = std_min, std_max, noise_prob
        self.channel_std = torch.as_tensor(IMAGENET_STD) # (C,)

    def __call__(self, x):
        if self.std_high <= 0 or self.noise_prob <= 0:
            return x
        leading = x.shape[:-3]
        stds = torch.empty(leading, device=x.device, dtype=x.dtype).uniform_(
            self.std_low, self.std_high)
        if self.noise_prob < 1.0:
            mask = (torch.rand(leading, device=x.device) < self.noise_prob).to(x.dtype)
            stds = stds * mask
        per_frame_scale = stds.view(*leading, 1, 1, 1)
        channel_factor = (1.0 / self.channel_std.to(x.device, x.dtype)).view(
```

```
        *([1] * len(leading)), -1, 1, 1)
    scale = per_frame_scale * channel_factor # pixel-space std -> normalized
    return x + torch.randn_like(x) * scale
```

B.3 Evaluation protocol

Unperturbed evaluation uses no noise, 100 trajectories per seed × 3 seeds (42/43/44), totalling 300 trajectories per condition. The primary noisy evaluation adds Gaussian noise of std $\in \{0.05, 0.08\}$ to observation pixels while leaving the goal image unperturbed under the same 3-seed protocol. Observation+goal corruption is retained as an auxiliary full-visual-stream stress condition and is reported separately where used. Success criterion is task-specific (PushT: T-block pose match; TwoRoom: target region; Reacher: joint-angle match; Cube: cube position match).

B.4 Full LeWM noise-sweep tables

Tables 8 and 9 give the complete LeWM sweep behind Table 2 and fig. 2. All cells are success rate mean ± population std across the three evaluation seeds.

Table 8: LeWM+noise sweep, unperturbed success rate (%).

$\sigma_{\max}$	TwoRoom	PushT	Reacher	Cube
0 (base)	94.00 ± 3.56	86.33 ± 2.36	58.67 ± 1.25	66.67 ± 2.62
0.01	93.67 ± 3.30	88.00 ± 3.74	61.67 ± 2.49	69.33 ± 0.47
0.02	95.00 ± 2.83	88.33 ± 2.62	85.67 ± 2.49	60.00 ± 1.63
0.03	96.33 ± 3.30	89.67 ± 1.70	78.67 ± 1.25	65.00 ± 1.63
0.04	96.33 ± 2.05	89.33 ± 2.05	84.00 ± 2.94	69.00 ± 3.74
0.05	96.00 ± 2.83	80.67 ± 4.78	70.00 ± 2.16	59.33 ± 0.94
0.06	96.67 ± 2.05	89.33 ± 2.05	86.00 ± 2.94	66.67 ± 2.05
0.07	96.00 ± 1.63	85.67 ± 3.09	83.67 ± 3.30	67.67 ± 0.94
0.08	98.33 ± 0.47	88.33 ± 2.87	84.00 ± 0.82	62.33 ± 1.25

Table 9: LeWM+noise sweep, observation-noise 0.08 success rate with unperturbed goal (%).

$\sigma_{\max}$	TwoRoom	PushT	Reacher	Cube
0 (base)	65.67 ± 1.25	4.33 ± 1.25	18.33 ± 2.05	47.00 ± 6.38
0.01	94.00 ± 2.83	51.67 ± 5.44	45.67 ± 3.86	49.00 ± 2.45
0.02	94.33 ± 3.09	73.67 ± 4.99	83.67 ± 2.49	58.67 ± 3.30
0.03	97.00 ± 2.94	82.00 ± 6.48	78.67 ± 2.87	68.33 ± 2.49
0.04	96.67 ± 2.05	86.67 ± 2.87	82.00 ± 0.82	65.00 ± 2.45
0.05	95.33 ± 3.30	78.33 ± 5.91	70.00 ± 2.94	60.00 ± 0.82
0.06	96.33 ± 2.05	87.67 ± 3.30	81.33 ± 1.70	66.67 ± 1.70
0.07	96.67 ± 1.25	84.00 ± 3.74	84.67 ± 5.19	67.33 ± 2.05
0.08	97.67 ± 0.94	89.00 ± 4.32	82.00 ± 1.41	62.00 ± 1.41

B.5 Full LeWM ACPC-basin grid

Table 3 is a compact baseline-vs-selected-endpoint summary. Table 10 lists the full 9-level LeWM grid from the released ACPC-basin artifact. The table is included to make the post-hoc nature of the basin probe explicit: R_F generally shrinks with input-side noise on the tasks with large cliffs, but it is not used to select checkpoints and is not a reliable standalone predictor of success.

Table 10: Full LeWM Gaussian-noise ACPC-basin grid. Success is observation-noise 0.08 success with an unperturbed goal; drop is unperturbed success minus that success. R_E and R_F are the normalised encoder-history and $H = 8$ rollout-view radii used in Table 3.

Task	$\sigma_{\max}$	unpert.	obs 0.08	drop	R_E	R_F	R_F/R_E	note
TwoRoom	0	94.00	65.67	28.33	3.538	0.785	0.222	base
TwoRoom	0.01	93.67	94.00	-0.33	1.633	0.232	0.142	
TwoRoom	0.02	95.00	94.33	0.67	0.931	0.124	0.133	
TwoRoom	0.03	96.33	97.00	-0.67	0.720	0.084	0.117	
TwoRoom	0.04	96.33	96.67	-0.33	0.518	0.056	0.108	
TwoRoom	0.05	96.00	95.33	0.67	0.360	0.039	0.108	
TwoRoom	0.06	96.67	96.33	0.33	0.300	0.034	0.112	
TwoRoom	0.07	96.00	96.67	-0.67	0.259	0.029	0.112	
TwoRoom	0.08	98.33	97.67	0.67	0.235	0.028	0.121	obs-best
PushT	0	86.33	4.33	82.00	1.727	1.543	0.893	base
PushT	0.01	88.00	51.67	36.33	0.677	0.474	0.701	
PushT	0.02	88.33	73.67	14.67	0.383	0.250	0.654	
PushT	0.03	89.67	82.00	7.67	0.253	0.177	0.699	
PushT	0.04	89.33	86.67	2.67	0.222	0.145	0.653	
PushT	0.05	80.67	78.33	2.33	0.232	0.141	0.607	
PushT	0.06	89.33	87.67	1.67	0.169	0.106	0.628	
PushT	0.07	85.67	84.00	1.67	0.159	0.105	0.663	
PushT	0.08	88.33	89.00	-0.67	0.137	0.088	0.642	obs-best
Reacher	0	58.67	18.33	40.33	4.238	1.012	0.239	base
Reacher	0.01	61.67	45.67	16.00	1.359	0.241	0.177	
Reacher	0.02	85.67	83.67	2.00	0.067	0.017	0.251	
Reacher	0.03	78.67	78.67	0.00	0.283	0.052	0.185	
Reacher	0.04	84.00	82.00	2.00	0.200	0.038	0.191	
Reacher	0.05	70.00	70.00	0.00	0.064	0.015	0.241	
Reacher	0.06	86.00	81.33	4.67	0.147	0.031	0.209	
Reacher	0.07	83.67	84.67	-1.00	0.135	0.027	0.200	obs-best
Reacher	0.08	84.00	82.00	2.00	0.116	0.024	0.209	
Cube	0	66.67	47.00	19.67	1.343	0.957	0.713	base
Cube	0.01	69.33	49.00	20.33	0.772	0.471	0.610	
Cube	0.02	60.00	58.67	1.33	0.063	0.019	0.293	
Cube	0.03	65.00	68.33	-3.33	0.238	0.115	0.481	obs-best
Cube	0.04	69.00	65.00	4.00	0.177	0.078	0.440	
Cube	0.05	59.33	60.00	-0.67	0.045	0.010	0.232	
Cube	0.06	66.67	66.67	0.00	0.120	0.046	0.385	
Cube	0.07	67.67	67.33	0.33	0.098	0.038	0.390	
Cube	0.08	62.33	62.00	0.33	0.094	0.036	0.385	

B.6 Compute

Training runs on a single NVIDIA H800 (80 GB) and follows the LeWM training schedule released with the baseline.

B.7 Main-figure rendering

Figures 2 and 6 are produced by the main plot script listed in Appendix B.10; no checkpoint or model loading is required.

Figure 3 is produced by `tools/paper1_selective_contraction.py`. The rendering command used for this paper is:

```
python -m tools.paper1_selective_contraction \
  --plot-clusters --plot-tasks PushT \
  --n-sequences 128 --cluster-anchor-count 16 \
  --view-stds 0.0 0.01 0.04 0.08 \
  --cluster-perturb-repeats 6 \
  --cluster-out-dir assets/paper1_figs \
  --cluster-envelope ellipse --cluster-envelope-coverage 0.90
```

Unlike the aggregate figure generator, this optional rendering path loads the PushT checkpoints and dataset windows to draw repeated same-state perturbation samples. The small panel summaries report high-dimensional statistics; the t-SNE ellipses are only qualitative 2-D covariance envelopes.

B.8 Local cluster atlas: a projection-free check of the basin visualization

Because t-SNE distorts global distances, Figure 4 provides a complementary *local normalized atlas* view of the same PushT data. Each small box selects one original dataset state; the circle radius equals the distance to its nearest *other* original state, computed in the original high-dimensional feature space, and the coloured markers are that state's original and Gaussian-perturbed views after a local PCA used only for in-box display. The visual question per box is therefore exactly the high-dimensional one: does the same-state perturbation cloud stay inside the nearest-different-state radius? On the no-noise-trained checkpoint many perturbation views escape their circles (median $r/\mathrm{NN} > 1$, few disjoint balls); on the full-sequence noise-trained checkpoint the clouds contract well inside the circles in both encoder and predictor-rollout features. This avoids the global-projection caveats of Figure 3 at the cost of showing only local neighbourhoods; the quantitative multi-task evidence remains Table 3.

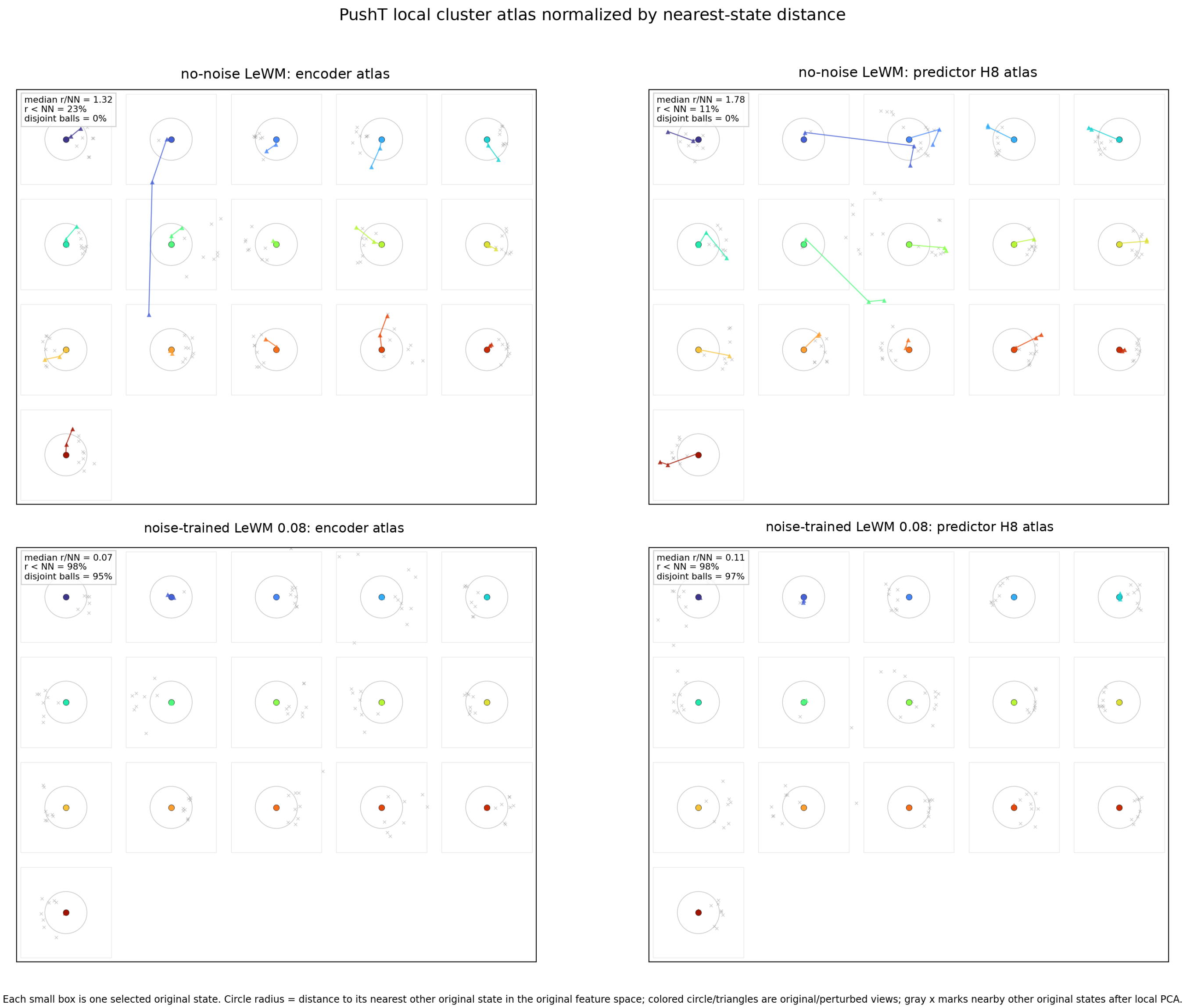


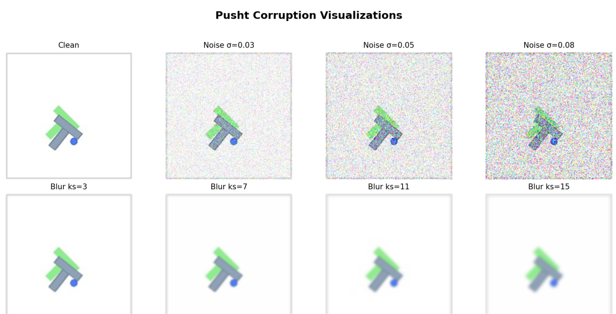
Figure 4: PushT local cluster atlas ($n = 128$ windows, 16 anchors, 8 neighbours). Each box is one original state with Gaussian-perturbed views after local PCA; the circle radius is the high-dimensional nearest-other-state distance. Panel summaries report original-space local statistics.

The atlas rendering command is:

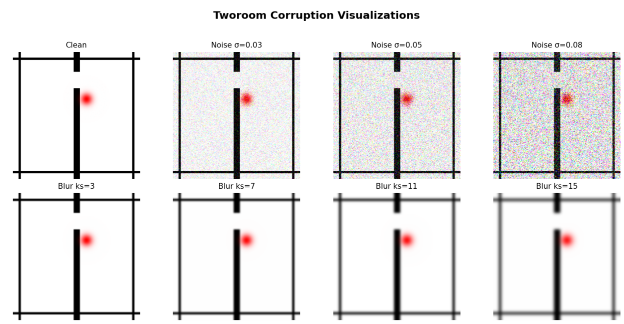
```
python -m tools.paper1_selective_contraction \
  --plot-atlas --plot-tasks PushT \
  --n-sequences 128 --atlas-anchor-count 16 \
  --view-stds 0.0 0.01 0.04 0.08 \
  --atlas-out-dir assets/paper1_figs
```

B.9 Corruption visualizations

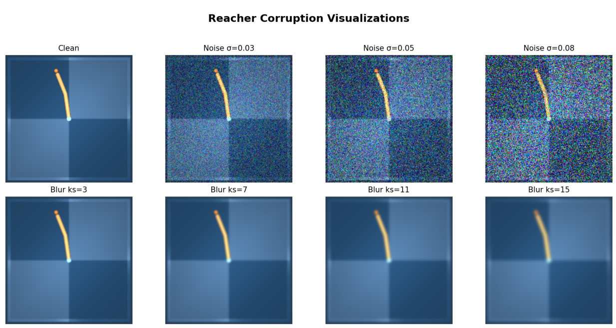
Figure 5 shows representative renderings of the two evaluated corruption families, applied to a sample observation from each of the four tasks. The Gaussian-noise row includes a mild calibration severity $\sigma = 0.03$ in addition to the two evaluation endpoints $\sigma \in \{0.05, 0.08\}$ used in Sections 5.1 and 5.2; the main sweep treats observation-only noise with an unperturbed goal as the primary endpoint and keeps observation+goal noise as an auxiliary stress condition. The Gaussian-blur kernel sizes ($k \in \{3, 7, 11, 15\}$) match the stress test in Section J. Each panel is a single 224×224 RGB frame from the corresponding task's evaluation rollout, after corruption and before encoder normalisation.



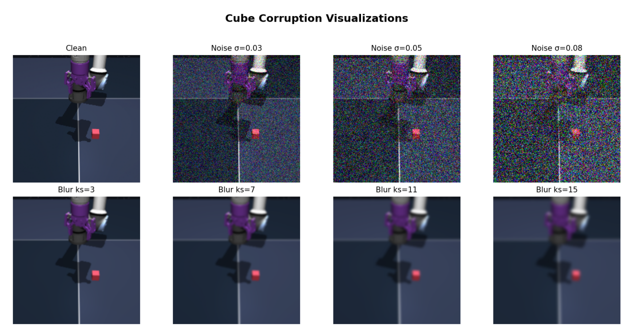
(a) PushT



(b) TwoRoom



(c) Reacher



(d) Cube

Figure 5: Corruption visualizations on the four tasks. Each panel is a 2×4 grid: additive Gaussian pixel noise on row 1 (Original, $\sigma \in \{0.03, 0.05, 0.08\}$) and Gaussian blur on row 2 ($k \in \{3, 7, 11, 15\}$, auto sigma). The $\sigma = 0.08$ endpoint is the primary Gaussian stress condition used in Tables 1 and 9.

B.10 Artifact and key-name mapping

The main text uses reader-facing diagnostic names. Exact relative artifact paths and implementation keys are collected here so that the manuscript remains readable while preserving reproducibility.

Table 11: Mapping from paper names to released relative paths and implementation keys.

Paper name	Released relative path or key
training-noise ceiling $\sigma_{\max}$	`std_max`
canonical evaluation artifact	`assets/paper1_data/canonical_evals_20260517.json`
canonical diagnostic artifact	`assets/paper1_data/canonical_diagnostics_20260517.json`
partial-correlation bootstrap artifact	`assets/paper1_data/partial_corr_bootstrap_20260523.json`
LeWM ACPC-basin artifact	`assets/paper1_data/acpc_basin_diagnostics.json`
PLDM ACPC-basin artifact	`assets/paper1_data/acpc_basin_diagnostics_pldm.json`
Phase-0 paired ACPC artifact	`assets/paper1_data/acpc_phase0_diagnostics.json`
blur baseline artifact	`assets/paper1_data/canonical_blur_baselines_20260523.json`
target-view ablation artifact	`assets/paper1_data/target_view_closed_loop_summary.json`
PLDM full-diagnostic artifact	`assets/paper1_data/canonical_full_diagnostics_pldm_20260523.json`
encoder-shift ratio	`noise_to_nn_cos_ratio`
fragility ratio	`predictor_target_to_nn_cos_ratio_at_max_std`
transition-resolution ratio	`transition_resolution_ratio_{cos,l2}`
multi-step rollout drift	`predictor_rollout_T8_l2`
ID probe R^2	`id_probe_r2`
main plot script	`tools/paper1_figs.py`
ACPC basin script	`tools/paper1_acpc_basin.py`
selective-contraction rendering script	`tools/paper1_selective_contraction.py`

C Diagnostic framework definitions

C.1 LeWM training objective

LeWM [23] is trained with latent prediction and SIGReg regularisation:

$$\mathcal{L}_{\text{LeWM}} = \mathcal{L}_{\text{pred}} + \lambda \cdot \mathcal{L}_{\text{SIGReg}}\,, \tag{7}$$

where $\mathcal{L}_{\text{pred}}$ is the latent-space MSE between predicted and target representations. SIGReg projects each latent onto unit-norm random directions, computes the Epps–Pulley empirical-characteristic-function distance [7] between each projection and $\mathcal{N}(0,1)$, and aggregates with Gauss-window weights to prevent collapse without explicit BatchNorm. Inference uses CEM [20] for latent-space MPC [8].

C.2 Five-layer diagnostic framework

The diagnostic framework decomposes the JEPA world-model control pass into pixels $\rightarrow$ encoder f $\rightarrow$ latent z $\rightarrow$ predictor g $\rightarrow$ cost surface $\rightarrow$ planned actions. Layers 1–2 describe whether visual perturbations enter the latent and how they change latent geometry. Layer 3 measures how the predictor amplifies an encoder shift. Layer 4 injects noise directly into z to separate encoder effects from predictor and cost effects. Layer 5 checks task-relevant information retained in the latent. We use nearest-neighbour distance as a local latent scale primitive in the spirit of kNN out-of-distribution detection [31]; the scale-normalised ratios are introduced for this diagnostic study.

Minimal notation. Let $z_i = f(x_i)$ and $\tilde{z}_i = f(\tilde{x}_i)$ be unperturbed and noised latents for the same token. The distance, local nearest-neighbour (NN) scale, and three introduced ratios are

$$\begin{aligned} d_{\cos}(a,b) &= 1 - \frac{a^\top b}{\|a\|\,\|b\|}, \\ d_i^{\mathrm{NN}} &= \min_{j\neq i} d_{\cos}(z_i, z_j), \\ r_i^{\mathrm{enc}} &= \frac{d_{\cos}(z_i, \tilde{z}_i)}{d_i^{\mathrm{NN}} + \epsilon}, \\ r_i^{\mathrm{pred}} &= \frac{d_{\cos}\big(g(z)_i, g(\tilde{z})_i\big)}{d_i^{\mathrm{NN}} + \epsilon}, \\ R_d &= \frac{\mathrm{median}_t \; d(z_t, z_{t+1})}{\mathrm{median}_{(t,t')\in\mathcal{F}} \; d(z_t, z_{t'}) + \epsilon}. \end{aligned} \tag{8}$$

Here r_i^{enc} is the encoder-shift ratio, r_i^{pred} is the fragility ratio, and R_d is the transition-resolution ratio for either cosine or L2 distance d. $\mathcal{F}$ denotes random far pairs from different temporal locations, and $\epsilon = 10^{-8}$ is a numerical-stability constant.

Layer metrics. Layer 1 reports raw angular/L2 encoder shift, angular gain per unit noise, and the encoder-shift ratio in Equation (8). Layer 2 reports local neighbourhood scale, effective rank [27], and CKA [19]. Layer 3 reports fragility ratio and multi-step rollout L2 drift. Layer 4 reports latent-noise cost-surface slope and latent robust radius. Layer 5 reports the L2/cosine transition-resolution ratio and inverse-dynamics linear-probe R^2 [1]. Layers 1, 2, 4, and 5 are descriptive; Layer 3 supplies the two full-coverage cross-checkpoint metrics used in Section 5.5.

C.3 Cross-checkpoint validation protocol

For each task we compute Spearman ρ across the 9 LeWM checkpoints and partial Spearman conditioned on $\sigma_{\max}$. The partial step removes the sweep-level training-noise trend. We report 95% percentile bootstrap intervals ($B = 1000$, seed = 42) in the released artifact listed in Appendix B.10.

D Full diagnostic profile of LeWM-base on four tasks

Table 12 summarises every core diagnostic on the four LeWM-base checkpoints. Exact released keys and diagnostic families are mapped in Appendix B.10; the table itself uses reader-facing metric names.

Table 12: Full LeWM-base diagnostics across four tasks.

Layer / metric	TwoRoom	PushT	Reacher	Cube	Unit / note
Encoder geometry					
Local NN cosine distance	0.0449	0.2360	0.0633	0.1856	cosine distance
Pairwise cosine distance	0.9904	1.0228	1.0252	1.0193	pair-wise cos dist
Effective rank	47.60	76.42	61.04	73.25	effective rank
Noise sensitivity					
Median noise angle (@ std 0.05)	5.51°	1.33°	3.22°	1.40°	median angular shift
Encoder-shift ratio	0.1031	0.011	0.0249	0.016	noise/NN cos ratio
Robust radius	0.0142	0.0537	0.0142	0.0356	critical noise std
First high-risk std	>0.08	>0.08	>0.08	>0.08	first high-risk std
Angular gain per std	1085.8	284.8	831.7	327.0	°/std angular gain
Geometry label	balanced	robust	balanced	robust	geometry label
Task resolution					
transition-resolution ratio (L2)	0.7216	0.3015	0.3704	0.4847	L2 resolution ratio
transition-resolution ratio (cos)	0.5538	0.0868	0.1351	0.2347	cos resolution ratio
ID probe R^2	0.2889	0.7739	0.1621	0.6657	linear probe R^2
Minimum ID probe R^2	0.2599	0.6786	0.1366	0.0972	min probe R^2
LiDAR rank proxy	46.06	13.95	45.90	42.46	LiDAR rank proxy
Action effect					
Mean action-induced prediction shift	0.5329	0.1283	0.2518	0.2364	action-perturb mean shift
Action-shift correlation	0.2847	0.2873	0.4042	0.2559	shift $\times$ $\|a\|$ corr
Predictor stability					
target/NN ratio	1.51e-4	3.54e-6	2.67e-5	3.39e-6	max-std target/NN ratio
Multi-step rollout drift	18.62	18.65	15.17	20.20	T=8 rollout L2 drift
Latent noise					
Linear CKA at max std	0.1986	0.5536	0.3085	0.1814	CKA unperturbed vs noisy
Latent cost-surface slope	635.31	1.3886	599.45	0.6208	goal latent cost slope

E PushT detailed partial-correlation view

Table 13 gives the detailed PushT $n = 9$ view for fragility ratio. It separates residual checkpoint quality from gap prediction. Lower fragility ratio aligns with better unperturbed and noisy success after conditioning on $\sigma_{\max}$, but the apparent correlation with the corruption drop is mediated by $\sigma_{\max}$.

Table 13: Spearman ρ (unconditional) and partial Spearman ρ conditioned on $\sigma_{\max}$, for fragility ratio on the PushT $n = 9$ LeWM sweep under the unified 3-seed $\times$ 100 eval. CIs are percentile 95% intervals over $B = 1000$ bootstrap resamples (seed = 42). The $\sigma_{\max}$-only top rows are univariate sweep diagnostics with no metric/outcome pairing, so no CI is reported.

Quantity	Spearman ρ	95% bootstrap CI
$\rho(\sigma_{\max}, \text{metric})$	+0.83	—
$\rho(\sigma_{\max}, \text{unperturbed})$	0.00	—
$\rho(\sigma_{\max}, \text{obs-noise } 0.08)$	+0.88	—
$\rho(\sigma_{\max}, \text{corruption drop})$	−0.98	—
$\rho(\text{metric, unperturbed})$ unconditional	−0.33	$[-0.95, +0.62]$
$\rho(\text{metric, unperturbed}) \mid \sigma_{\max}$ partial	−0.59	$[-0.97, -0.10]$
$\rho(\text{metric, obs-noise } 0.08)$ unconditional	+0.60	$[-0.11, +1.00]$
$\rho(\text{metric, obs-noise } 0.08) \mid \sigma_{\max}$ partial	−0.53	$[-0.84, 0.00]$
$\rho(\text{metric, corruption drop})$ unconditional	−0.80	$[-1.00, -0.23]$
$\rho(\text{metric, corruption drop}) \mid \sigma_{\max}$ partial	+0.19	$[0.00, +0.70]$

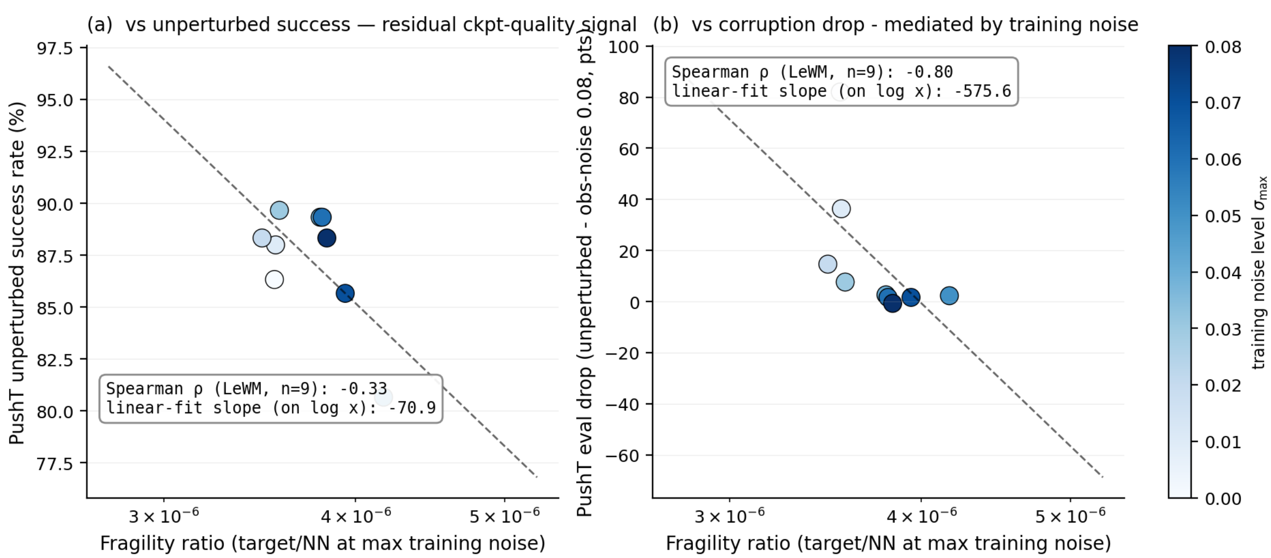


Figure 6: PushT $n = 9$ LeWM sweep: fragility ratio is a residual checkpoint-quality signal for unperturbed success (a); the apparent corruption-drop correlation in (b) is mediated by $\sigma_{\max}$ (encoded by the colour bar).

Panel (a) plots fragility ratio against unperturbed success. Panel (b) plots fragility ratio against corruption drop; the colour bar shows that low-$\sigma_{\max}$ checkpoints have high drop, high-$\sigma_{\max}$ checkpoints have low drop, and the metric tracks $\sigma_{\max}$ itself. After the $\sigma_{\max}$ trend is removed, the metric does not robustly separate small-drop from large-drop checkpoints.

F Heteroscedastic-loss formulation

The scale-preserving heteroscedastic NLL referenced in Section 6 and detailed in Section G is:

$$\mathcal{L}_{\text{hetero}} = \tfrac{1}{2}\, \exp(-s_t)\, \|z_{t+1} - \hat{z}_{t+1}\|^2 + \tfrac{1}{2}\, s_t\,, \tag{9}$$

where s_t is the σ-head-predicted log-variance. During training $\exp(-s_t)$ acts as an automatic weight: high-error transitions are down-weighted and low-error transitions are up-weighted. When $s_t \equiv 0$ the loss reduces to plain

MSE.

G Heteroscedastic-loss negative result (data)

This appendix records the full heteroscedastic-loss data referenced in Section 6. The σ-head is trained jointly with the predictor; gradient is detached on the σ path so the mean prediction path is exactly LeWM MSE when σ is constant.

Table 14: Heteroscedastic-loss evaluation (canonical 3-seed × 100 protocol). The LeWM+noise rows are each task's observation+goal 0.08 point-best checkpoint (TwoRoom $\sigma_{\max} = 0.08$, PushT $\sigma_{\max} = 0.06$), matching the strongest stress column of this ablation; the observation-only point-bests in Table 9 differ.

Task / model	unpert.	goal .05	obs .05	obs+goal .05	goal .08	obs+goal .08
TwoRoom LeWM-base	94.00	73.33	72.33	61.33	58.67	50.00
TwoRoom LeWM+noise point-best	98.33	98.00	98.33	98.00	98.67	98.67
TwoRoom hetero	99.67	85.33	96.67	84.67	73.33	55.33
PushT LeWM-base	86.33	38.00	17.00	12.00	11.33	4.67
PushT LeWM+noise point-best	89.33	87.67	88.00	88.33	89.67	87.00
PushT hetero	13.33	7.67	7.67	7.67	9.67	6.00

TwoRoom hetero reaches 99.67% on unperturbed evaluation — compatible with the prior that low-dimensional discrete tasks can tolerate stronger nuisance contraction / clustering — but lags noise training on high-noise robustness. **PushT hetero unperturbed success is** 13.33% — **a method-level failure**, not evidence for a useful robustness compromise.

Table 15: Representation diagnostics of the hetero-loss failure.

Metric	TwoRoom base	TwoRoom hetero	PushT base	PushT hetero
Local NN cosine distance	0.0449	0.0281	0.2360	0.1051
Effective rank	47.60	33.59	76.42	42.85
transition-resolution ratio (cos)	0.5538	0.3780	0.0868	0.0101
transition-resolution ratio (L2)	0.7216	0.6055	0.3015	0.1023
ID probe R^2	0.2889	−0.0573	0.7739	0.2678
Mean action-induced prediction shift	0.5329	0.4482	0.1283	0.0841
Multi-step rollout drift	18.62	17.90	18.65	14.01

Hetero-loss compresses the representation in both tasks. In TwoRoom (low-dimensional, discrete, redundant), the observed compression coexists with high unperturbed success, but the robustness shortfall shows that compression alone is not the target. For PushT, the L2 transition-resolution ratio collapses from 0.3015 to 0.1023 and ID probe R^2 from 0.7739 to 0.2678 — task-relevant state information is erased. The contrast with input-side noise training is sharp: the PushT noise-sweep representative in Table 4 leaves rank and probe nearly unchanged, so this collapse is specific to error-based downweighting. The drop in multi-step rollout drift is not good news either: the latent has become more *predictable* without being more *controllable*.

Mechanism. The learned σ values align with visible high-error transition regimes (contact frames in PushT, doorway crossings in TwoRoom, etc.), but using σ as an automatic weight to down-weight high-error transitions misclassifies the contact-and-control-critical states of PushT as "unimportant" and erases them. This is a negative example of the broader selective-consistency lesson: in contact-heavy control, **hard ≠ unimportant**.

This finding motivates a narrower follow-up direction: per-token *consistency* controllers should be separated from loss reweighting, and any difficulty signal should be detached from the mean prediction path unless it is explicitly protected by an action-relevance guard. We leave that objective design outside the present diagnostic study.

H Mechanism-attribution probe

Section 5.4 reports what moves in the representation and Section 5.5 reports which diagnostics correlate with evaluation across checkpoints. To localise whether the perturbation enters through the encoder or later predictor/planner stages, we use latent-noise probing as a conservative mechanism check. Directly injecting noise into latent z skips the encoder and decouples encoder contributions from predictor and cost contributions.

Table 16: Three-layer attribution by task (LeWM $n = 9$ sweep, unified 3-seed $\times$ 100 eval).

Task	Strongest signal in these probes	Residual after removing the $\sigma_{\max}$ trend
TwoRoom	encoder-shift signal (rank-saturated)	near-zero residual; saturation limits inference
PushT	encoder + single-step predictor (both mediated by $\sigma_{\max}$)	no residual metric isolates corruption drop
Reacher	encoder + multi-step rollout	weak multi-step drift residual signal (partial $\rho = +0.37$)
Cube	encoder	weak residual on multi-step drift (partial $\rho = +0.23$)

Across these probes, the strongest common signal is encoder-side perturbation transduced by the predictor. PushT is the clearest mediated case: multi-step rollout drift has unconditional $\rho = +0.93$ against corruption drop, but partial $\rho = -0.01$ after removing the $\sigma_{\max}$ trend; the single-step fragility ratio similarly changes from unconditional $\rho = -0.80$ to partial $\rho = +0.19$. Reacher and Cube retain only weak residual multi-step signals, and TwoRoom is limited by saturation. A stronger causal claim about planner or cost-surface contributions would require multi-task latent-injection or cost-ranking ablations.

I Cross-method replication: PLDM noise sweep on four tasks

This appendix records the second method-family on which the noise sweep and the cross-checkpoint correlation analysis replicate. PLDM follows the joint-embedding predictive line from Sobal et al. [29]; the concrete latent-dynamics planning baseline is from Sobal et al. [30]. We use the implementation distributed through the `stable-worldmodel` baseline suite [22]. Training and evaluation follow the LeWM canonical protocol bit-for-bit: per-frame Gaussian pixel noise, $\sigma_{\max} \in \{0.01, \ldots, 0.08\}$, three evaluation seeds (42/43/44), 100 trajectories per seed.

Coverage. The PLDM release is complete for the same grid as the LeWM canonical sweep: 4 tasks $\times$ 9 configurations (no-noise baseline plus $\sigma_{\max} \in \{0.01, \ldots, 0.08\}$), three evaluation seeds (42/43/44), and 100 trajectories per seed. The PLDM eval, predictor-diagnostic, full-diagnostic, and cross-method-correlation JSON files are listed in the data manifest; we do *not* merge them into the LeWM canonical files used by Tables 1–13.

No-noise-trained PLDM cliff. PLDM reproduces the no-noise-trained visual-corruption cliff on PushT and TwoRoom, while Reacher and Cube show much smaller drops under this PLDM training recipe (Table 17). This supports a method-family reading of the failure mode without claiming every task has the same cliff magnitude.

Table 17: PLDM baseline trained without input-noise augmentation on all four tasks. Success rate mean $\pm$ population std, 3 seeds $\times$ 100 trajectories.

Task	unpert.	obs 0.05	obs 0.08	obs+goal 0.08	drop
TwoRoom	96.67 ± 1.70	79.67 ± 4.03	67.00 ± 2.16	51.67 ± 2.05	-29.67
PushT	75.33 ± 3.68	46.00 ± 4.97	18.33 ± 2.05	10.00 ± 2.16	-57.00
Reacher	82.67 ± 1.70	81.33 ± 4.03	80.33 ± 5.73	79.33 ± 0.94	-2.33
Cube	52.67 ± 3.30	50.33 ± 4.50	48.33 ± 4.50	48.33 ± 2.87	-4.33

PLDM full sweep — unperturbed evaluation. Table 18 reports the per-task unperturbed success rate at every trained $\sigma_{\max}$ level. Table 19 reports the corresponding observation-noise 0.08 robustness with an unperturbed goal.

Table 18: PLDM+noise sweep, unperturbed success rate (%). † marks the per-task unperturbed point-best.

$\sigma_{\max}$	TwoRoom	PushT	Reacher	Cube
0 (base)	96.67 ± 1.70	75.33 ± 3.68	82.67 ± 1.70	52.67 ± 3.30
0.01	96.33 ± 0.94	72.33 ± 4.19	78.00 ± 0.82	51.33 ± 1.25
0.02	97.33 ± 0.47	71.33 ± 3.86	82.33 ± 2.36	53.33 ± 1.70
0.03	96.67 ± 1.70	$76.67 \pm 7.54^{\dagger}$	83.00 ± 3.74	52.67 ± 3.30
0.04	97.67 ± 0.47	68.67 ± 5.25	$85.00 \pm 2.16^{\dagger}$	54.00 ± 2.94
0.05	97.67 ± 1.25	74.33 ± 3.40	72.00 ± 1.41	54.00 ± 2.16
0.06	98.00 ± 0.82	70.33 ± 6.18	79.00 ± 0.82	$56.00 \pm 4.97^{\dagger}$
0.07	98.00 ± 0.82	66.67 ± 8.38	80.67 ± 2.62	53.67 ± 3.68
0.08	$98.33 \pm 0.94^{\dagger}$	68.00 ± 1.41	80.33 ± 1.25	54.00 ± 1.63

Table 19: PLDM+noise sweep, observation-noise 0.08 success rate with unperturbed goal (%). ‡ marks the per-task observation-noise point-best.

$\sigma_{\max}$	TwoRoom	PushT	Reacher	Cube
0 (base)	67.00 ± 2.16	18.33 ± 2.05	80.33 ± 5.73	48.33 ± 4.50
0.01	96.00 ± 1.63	23.33 ± 2.49	77.67 ± 3.40	51.33 ± 2.36
0.02	95.67 ± 0.94	47.00 ± 0.82	80.67 ± 5.25	51.00 ± 4.32
0.03	96.33 ± 1.25	$73.00 \pm 7.48^{\ddagger}$	$81.67 \pm 4.71^{\ddagger}$	54.33 ± 2.62
0.04	97.33 ± 0.94	66.67 ± 5.73	80.33 ± 2.87	$54.67 \pm 2.36^{\ddagger}$
0.05	97.67 ± 1.89	71.67 ± 5.79	62.67 ± 4.19	54.00 ± 3.74
0.06	$98.00 \pm 0.00^{\ddagger}$	69.33 ± 4.78	75.00 ± 0.82	54.33 ± 0.94
0.07	98.00 ± 0.82	64.00 ± 7.79	78.33 ± 2.05	53.00 ± 3.74
0.08	97.67 ± 1.25	68.00 ± 5.66	79.33 ± 1.25	53.67 ± 2.05

Reading. Four quantitative observations are useful for interpreting PLDM as a second method family:

1. **TwoRoom (visually redundant) shows high-noise recovery on both methods.** PLDM observation-noise 0.08 success rises from 67.00% at the no-noise baseline to 98.00% at $\sigma_{\max} = 0.06$, then remains on a high-noise plateau. This mirrors the LeWM reading that TwoRoom tolerates strong same-state nuisance contraction.
2. **PushT (contact-heavy) exhibits the corruption cliff and large noise-training recovery.** PLDM without noise augmentation drops by 57.00 pt under observation-noise 0.08; training with $\sigma_{\max} = 0.03$ recovers observation-noise 0.08 success to 73.00% (+54.67 pt over the no-noise baseline). The unperturbed and observation-noise 0.08 point-bests are co-located at $\sigma_{\max} = 0.03$ on PLDM.
3. **Reacher is already robust under no-noise-trained PLDM.** The no-noise PLDM drop is only 2.33 pt, and the observation-noise point-best sits at $\sigma_{\max} = 0.03$ (81.67%). We therefore treat Reacher as a low-cliff external baseline rather than evidence for a universal noise-training gain.
4. **Cube (structured 3D manipulation) remains weakly noise-responsive.** PLDM Cube unperturbed success spans 51.33–56.00% and observation-noise 0.08 spans 48.33–54.67% across the full grid. This is the same weak noise responsiveness seen on LeWM.

Method-specific details. PLDM's PushT unperturbed point-best (76.67% at $\sigma_{\max} = 0.03$) is lower than LeWM's (89.67% at $\sigma_{\max} = 0.03$), so the external baseline is not a performance leaderboard. It is a robustness check: the cliff-and-recovery shape is reproduced, while the exact unperturbed/robust point-best location is method-dependent. In particular, the LeWM PushT point-best split should be read as a LeWM-family signature, not a cross-method law.

Cross-method partial correlations. We repeat the fragility ratio partial-correlation analysis for PLDM on all four tasks. The headline PushT null replicates in the limited sense that we do not see stable evidence for a non-zero residual corruption-gap association: PLDM has unconditional $\rho(\text{metric}, \text{corruption drop}) = -0.75$ but partial $\rho(\text{metric}, \text{corruption drop}) \mid \sigma_{\max} = -0.05$ (95% bootstrap CI $[-0.92, +0.61]$), and the joint LeWM+PLDM PushT partial after conditioning on $\sigma_{\max}$ and method is $+0.22$ (95% CI $[-0.59, +0.61]$). The intervals are wide and include zero. Other tasks show task-specific residuals, so the full table is evidence for a bounded diagnostic reading rather than a general robustness predictor.

Table 20: PLDM fragility ratio partial correlations, $n = 9$ per task. The first three partial columns condition on $\sigma_{\max}$ within PLDM; the final column uses the joint LeWM+PLDM sample ($n = 18$) and conditions on both $\sigma_{\max}$ and a method dummy.

Task	n	unpert.	obs 0.08	drop	joint drop
TwoRoom	9	−0.26	−0.43	+0.35	+0.12
PushT	9	−0.22	−0.13	−0.05	+0.22
Reacher	9	−0.68	−0.49	+0.17	+0.43
Cube	9	−0.36	+0.19	−0.53	−0.13

PLDM full-sweep ACPC basin replication. We also run the Gaussian-noise basin protocol on all 36 PLDM checkpoints. Table 21 reports the compact baseline-vs-observation-noise-0.08 point-best summary; the full 4 tasks × 9 configurations grid is listed in Appendix B.10.

Table 21: PLDM full-sweep Gaussian-noise ACPC basin replication, summarised by the no-noise baseline and each task's PLDM observation-noise 0.08 point-best checkpoint. The protocol matches Table 3: same-state Gaussian-noise views of the observation history (std 0.01–0.08), unperturbed goal, and the same recorded action sequence.

Task	checkpoint	obs 0.08	drop	R_E	R_F
TwoRoom	base	67.00	29.67	3.287	0.611
TwoRoom	noise 0.06	98.00	0.00	0.235	0.034
PushT	base	18.33	57.00	0.874	0.846
PushT	noise 0.03	73.00	3.67	0.153	0.091
Reacher	base	80.33	2.33	0.237	0.060
Reacher	noise 0.03	81.67	1.33	0.135	0.039
Cube	base	48.33	4.33	0.831	0.453
Cube	noise 0.04	54.67	−0.67	0.146	0.054

Reading. The PLDM basin replication is directionally consistent with the LeWM ACPC-basin reading: the observation-noise 0.08 point-best checkpoint reduces R_F on all four tasks, sharply on TwoRoom, PushT, and Cube. Reacher is the boundary case: PLDM is already near-robust at the no-noise baseline, and both the behavioural drop and the basin change are small. The full 36-row artifact reduces the concern that the paired basin pattern is purely a LeWM artifact, but it does not upgrade the claim to a general mechanism because Table 22 shows different internal diagnostic movement.

PLDM full-diagnostic mechanism check. We also run the same five diagnostic layers on the PLDM sweep. The compact base-vs-representative view in Table 22 matches the dominant LeWM movement in Table 4: representative recovery checkpoints on both families preserve rank, transition resolution, and the inverse-dynamics probe to first order, while strongly reducing multi-step predictor drift. The clearest method-specific difference is TwoRoom: LeWM's heavy-noise representative also compresses effective rank (47.6 → 37.7), whereas PLDM's representative leaves rank essentially unchanged (74.8 → 72.7). This separates two claims: the task-level fragility/recovery signature replicates across method families, while the residual diagnostic movement remains architecture-dependent.

Table 22: PLDM full-diagnostic check, no-noise baseline → representative noise-trained checkpoint. Representatives are each task's PLDM observation-noise 0.08 point-best checkpoint (TwoRoom $\sigma_{\max} = 0.06$, PushT $\sigma_{\max} = 0.03$, Reacher $\sigma_{\max} = 0.03$, Cube $\sigma_{\max} = 0.04$).

Task	rep. $\sigma_{\max}$	eff. rank	trans. L2	ID probe R^2	rollout T8 L2
TwoRoom	0.06	74.78 → 72.70	0.8188 → 0.8254	0.3233 → 0.3201	20.36 → 1.02
PushT	0.03	130.13 → 131.90	0.4710 → 0.4778	0.7570 → 0.7479	20.25 → 2.82
Reacher	0.03	117.52 → 108.14	0.5053 → 0.4824	0.2150 → 0.1844	1.45 → 0.94
Cube	0.04	134.80 → 124.49	0.6395 → 0.6313	0.5428 → 0.5576	18.98 → 4.02

What this strengthens, and what it does not. PLDM reduces the concern that the four task signatures are tied to one LeWM checkpoint family. The full PLDM basin artifact, summarised in Table 21, further

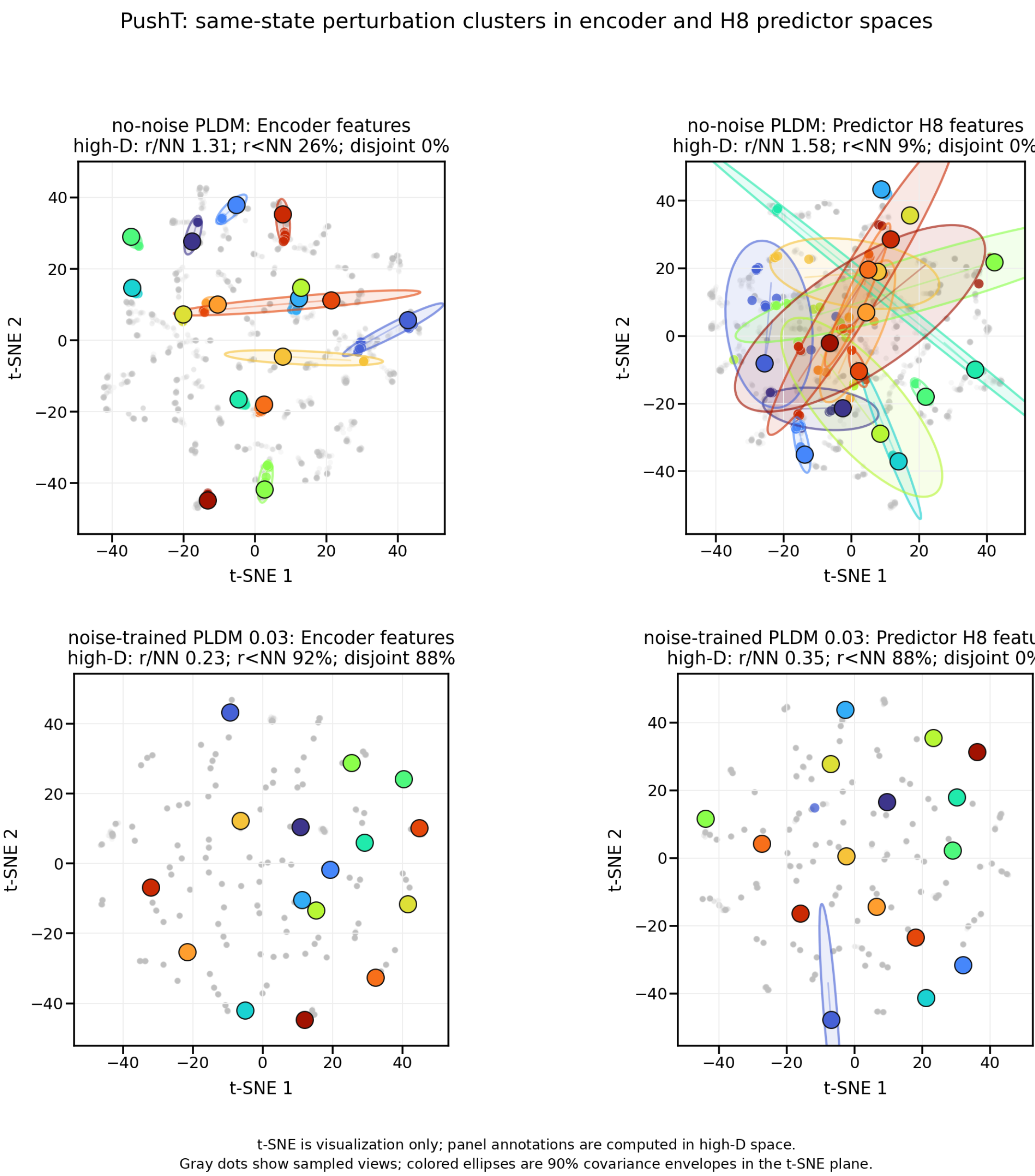


Figure 7: Qualitative PLDM PushT same-state perturbation clusters for Table 21. Top: no-noise-trained PLDM; bottom: noise-trained PLDM at $\sigma_{\max} = 0.03$. Small panel summaries report original-space local statistics; t-SNE coordinates are qualitative and not comparable with LeWM.

supports the paired same-state perturbation reading at the baseline-vs-robust endpoints. It also supports the diagnostic-boundary claim: after removing the sweep-level $\sigma_{\max}$ effect and the method offset on PushT, the local predictor-sensitivity diagnostic does not isolate corruption-specific robustness. The TwoRoom and Cube PLDM residuals remain visible in Table 20. They do not contradict the headline null: the intervals are wide, the noisy-eval ranges are small on the saturated tasks, and the residuals should not be converted into a selection rule.

Mechanism boundary. Table 22 is the reason we do not promote any single internal mechanism to a general explanation. On both families the most consistent representative-checkpoint movement is a strong reduction in multi-step rollout drift, but the residual movement is method-specific: LeWM compresses TwoRoom's effective rank where PLDM does not, and the two families place their weak residual partial-correlation signals on different tasks (Tables 6 and 20). The shared conclusion is therefore about *control-time visual fragility and noise-training recovery*; the internal diagnostic route is method-specific.

J Cross-corruption sanity check: no-noise-trained blur evaluation

This appendix addresses the narrow concern that the observed visual fragility is an artefact of Gaussian noise only. We evaluate the LeWM and PLDM baselines trained without input-noise augmentation under Gaussian blur with kernel sizes $3, 7, 11, 15$, applied to pixels, goal images, or both. This is an **eval-only** stress test: no blur-trained checkpoint is included, and the blur data are not mixed into the Gaussian-noise sweep. The released source is listed in Appendix B.10.

Table 23: Observation+goal blur stress test on baselines trained without input-noise augmentation. Success rate mean ± population std over 3 evaluation seeds × 100 trajectories. Kernel sizes 11 and 15 are severe low-pass endpoints on 224×224 inputs, so they should not be read as mild corruptions.

Method	Task	unpert.	$k = 3$	$k = 7$	$k = 11$	$k = 15$	worst drop
LeWM	TwoRoom	94.00 ± 3.56	39.33 ± 2.05	32.67 ± 3.09	30.33 ± 2.87	30.67 ± 0.47	-63.67
LeWM	PushT	86.33 ± 2.36	81.67 ± 4.50	74.00 ± 1.41	65.33 ± 2.49	58.67 ± 6.65	-27.67
LeWM	Reacher	58.67 ± 1.25	57.00 ± 6.38	45.67 ± 2.05	36.00 ± 4.90	20.33 ± 2.49	-38.33
LeWM	Cube	66.67 ± 2.62	65.00 ± 1.63	62.33 ± 2.62	57.33 ± 2.87	54.00 ± 2.16	-12.67
PLDM	TwoRoom	96.67 ± 1.70	38.33 ± 0.47	30.33 ± 2.87	28.33 ± 1.25	28.33 ± 1.70	-68.33
PLDM	PushT	75.33 ± 3.68	74.00 ± 2.94	72.00 ± 3.56	67.67 ± 4.03	62.33 ± 2.05	-13.00
PLDM	Reacher	82.67 ± 1.70	86.00 ± 2.16	80.00 ± 3.56	72.00 ± 2.16	68.00 ± 4.08	-14.67
PLDM	Cube	52.67 ± 3.30	53.00 ± 5.35	53.00 ± 2.83	50.67 ± 3.68	52.33 ± 5.44	-2.00

Reading. Blur is a different failure mode from Gaussian pixel noise. The clear collapse is TwoRoom: it falls below 40% already at $k = 3$ on both LeWM and PLDM and then saturates around 28–33% for $k = 7$–15. This reverses the Gaussian-noise ordering, where PushT is the most fragile task. The other tasks are more stable under blur: PLDM PushT, PLDM Reacher, and both Cube baselines lose at most about 15 pt at the strongest endpoint, while LeWM PushT degrades gradually and LeWM Reacher drops sharply only at $k = 15$. We therefore use blur only as an eval-only cross-corruption sanity check: visual fragility is not unique to Gaussian noise, but the task ordering and recovery profile are corruption-specific.

K Phase-0 paired ACPC diagnostics

K.1 Exploratory readout definitions

The main text fixes ACPC to the $H = 8$ identity-readout rollout-latent basin diagnostic. The additional readouts below are exploratory and are not used for model selection.

Encoder perturbation difference (EPD).
: $\mathrm{EPD} = d(E(o), E(\tilde{o}))$. This reports whether a perturbation enters the latent, not whether it matters for control.

One-step consistency (ACPC-1).
: The one-step paired score is

$$\mathrm{ACPC}_1 = d\big(F(E(o), a), F(E(\tilde{o}), a)\big).$$

For cross-checkpoint comparisons we optionally divide by the natural transition magnitude $d(z_t, z_{t+1}) + \epsilon$.

Multi-step consistency (ACPC-H).
: Equation (3) with $H \in \{4, 8\}$ measures disagreement between unperturbed and corrupted rollouts under the same action sequence.

Predictive cost consistency (PCC).
: $\mathrm{PCC} = \big|J(F^H(E(o), \mathbf{a}), g) - J(F^H(E(\tilde{o}), \mathbf{a}), g)\big|$ for a cost/goal readout J and goal g. PCC tests whether predictive mismatch changes the planning cost.

Candidate-ranking agreement (CRA).
: For a shared candidate set $\mathcal{A} = \{\mathbf{a}^1, \ldots, \mathbf{a}^K\}$, CRA is the Spearman/Kendall agreement of the unperturbed and corrupted cost vectors.

Margin-conditioned action flip (MAF).
: $\mathrm{MAF} = \mathbf{1}\big[u^*_{\mathrm{orig}} \neq u^*_{\mathrm{noise}},\ C_{(2)} - C_{(1)} > \delta\big]$: a flip of the selected action counts only when the unperturbed top-1/top-2 cost margin exceeds δ.

Action-relevant discriminability margin (ADM).
: Over action/transition-distinct pairs P_{disc}, $\mathrm{ADM} = \mathrm{median}_{(i,j)\in P_{\mathrm{disc}}} d(\Psi_i, \Psi_j)$. In this paper ADM is a proxy guard rather than a direct proof of the margin condition in Equation (5).

Selective predictive robustness ratio (SPRR).
: A summary of action-distinct discriminability relative to original/corrupted predictive inconsistency,

$$\mathrm{SPRR} = \frac{\mathrm{median}_{(i,j)\in P_{\mathrm{disc}}} d(\Psi_i, \Psi_j)}{\mathrm{median}_{P_{\mathrm{pert}}} \mathrm{ACPC}_H + \epsilon}.$$

K.2 Phase-0 table

This appendix records the first full-coverage run of these paired action-conditioned diagnostics. The released artifact covers 72 checkpoint rows: LeWM and PLDM, four tasks, and nine training-noise levels per task. All rows completed successfully. Each row uses 100 sampled sequences, Gaussian observation+goal corruption at std 0.08, rollout horizon $H = 8$, and a shared candidate-action set of one dataset future plus 64 random futures. Because this uses the auxiliary observation+goal stress endpoint rather than the paper's primary unperturbed-goal endpoint, it is reported as a broader exploratory sanity check rather than as the main ACPC evidence. PCC, CRA, and MAF are computed from the same original/noisy candidate-cost vectors. ADM and SPRR use an action-distance latent proxy, not an oracle state/keypoint margin.

Table 24: Phase-0 paired ACPC diagnostics, no-noise baseline → observation+goal 0.08 point-best checkpoint. Lower is better for ACPC-H/transition, PCC, and MAF; higher is better for CRA and top-8 overlap. Values are exploratory diagnostics, not predictor-selection rules.

Task	Method	robust $\sigma_{\max}$	obs+goal	ACPC-H/trans.	PCC	CRA	top-8	MAF
TwoRoom	LeWM	0.08	50.00 → 98.67	1.254 → 0.042	176.6 → 7.0	0.157 → 0.990	0.201 → 0.952	0.92 → 0.06
TwoRoom	PLDM	0.05	51.67 → 98.33	1.062 → 0.063	164.3 → 7.1	0.367 → 0.982	0.287 → 0.911	0.85 → 0.05
PushT	LeWM	0.06	4.67 → 87.00	2.740 → 0.149	67.3 → 4.6	0.323 → 0.986	0.213 → 0.914	0.96 → 0.08
PushT	PLDM	0.03	10.00 → 72.00	1.682 → 0.162	52.0 → 8.1	0.300 → 0.967	0.286 → 0.869	0.90 → 0.08
Reacher	LeWM	0.02	15.00 → 85.67	1.922 → 0.023	528.0 → 1.2	0.242 → 0.999	0.219 → 0.990	0.97 → 0.01
Reacher	PLDM	0.04	79.33 → 81.33	0.104 → 0.063	11.7 → 6.3	0.965 → 0.987	0.930 → 0.949	0.13 → 0.10
Cube	LeWM	0.07	46.33 → 68.00	1.944 → 0.055	88.8 → 4.1	0.337 → 0.996	0.266 → 0.951	0.88 → 0.00
Cube	PLDM	0.08	48.33 → 56.33	1.136 → 0.034	113.6 → 3.0	0.663 → 0.997	0.512 → 0.967	0.74 → 0.01

Reading. The Phase-0 pass converts the ACPC definitions from a prospective protocol into a release artifact that behaves sensibly on the existing sweep. The largest Gaussian-noise cliffs (TwoRoom and PushT on both methods, Reacher/Cube on LeWM) coincide with large baseline ACPC-H, PCC, ranking disruption, and action flips, and the observation+goal point-best checkpoints reduce these quantities sharply. PLDM Reacher is the important boundary case: it already has high observation+goal success at the no-noise baseline, and the paired diagnostics are correspondingly already close to the recovered checkpoint. Across the joint LeWM+PLDM sample within each task, partial Spearman correlations after conditioning on $\sigma_{\max}$ and method remain task-specific: ACPC-H/transition vs. corruption drop is strong on PushT (+0.67) and Reacher (+0.71), but only modest on TwoRoom (+0.21) and Cube (+0.33). We therefore treat Table 24 as a face-validity and local-diagnostic check, not as evidence that ACPC-H, PCC, CRA, MAF, ADM, or SPRR can by themselves predict robustness.

L Target-view ablation completion check

This appendix records the completed LeWM target-view ablation referenced in Section 6. The runs use the origin-future target configuration: the prediction context is encoded from the configured Gaussian-perturbed history, while the prediction target remains the original future latent. This is the perturbed-history → original-future branch, contrasted with the default full-sequence perturbed-target branch used by the main LeWM+noise sweep. The released summary artifact is listed in Appendix B.10.

We verified the four-task run set after completion: eight target-noise checkpoints per task and epoch-10 summaries for unperturbed, observation-noise 0.03, observation-noise 0.05, and observation-noise 0.08 evaluation. The canonical table below uses seeds 42/43/44 and 100 trajectories per seed.

Table 25: LeWM perturbed-history → original-future target-view ablation. Values are success rate mean ± population std over the eight target-noise checkpoints; each checkpoint value is the three-seed mean from `eval_summary.csv`.

Task	unperturbed	obs 0.03	obs 0.05	obs 0.08
TwoRoom	93.33 ± 1.72	88.67 ± 6.04	77.71 ± 9.12	61.75 ± 7.65
PushT	82.92 ± 4.00	57.71 ± 8.96	20.21 ± 11.68	6.75 ± 5.00
Reacher	59.92 ± 1.80	40.38 ± 2.78	32.00 ± 4.57	19.62 ± 3.97
Cube	65.00 ± 2.01	61.42 ± 2.06	44.29 ± 1.32	39.62 ± 1.22

Table 26: Closed-loop target-view comparison at observation-noise 0.08. Values are success rate mean ± population std over the eight training-noise checkpoints; each checkpoint value is the canonical three-seed mean.

Task	full-sequence perturbed target	perturbed-history → original future	full-seq advantage
TwoRoom	94.71 ± 1.90	61.75 ± 7.65	+32.96
PushT	72.83 ± 12.84	6.75 ± 5.00	+66.08
Reacher	76.17 ± 10.89	19.62 ± 3.97	+56.54
Cube	59.83 ± 5.55	39.62 ± 1.22	+20.21

Reading. The ablation acts as a negative scope check and supports retaining the full-sequence empirical branch. Simply changing the prediction target to the original future does not reproduce the recovery from input-side noise augmentation in Table 9: PushT and Reacher remain close to their no-noise robustness cliffs, TwoRoom degrades under stronger eval noise despite high unperturbed success, and Cube does not improve over the main sweep.

A matched PushT 0to008 check shows the contrast: rerunning both branches' $\sigma_{\max} = 0.08$ checkpoints under the same fixed evaluation code and canonical seeds gives observation-noise 0.08 three-seed means of 86.33% for the full-sequence checkpoint versus 8.67% for the origin-target checkpoint (the canonical sweep value for the same full-sequence checkpoint in Table 9 is 89.00%; the difference is normal evaluation variance). We interpret this as evidence consistent with a closed-loop train/eval consistency explanation: full-sequence noise augmentation better matches the planner's latent autoregressive feedback, while one-step origin-target denoising does not. This is not evidence that arbitrary history mismatches are harmless; it says the target view alone is not the missing mechanism, and an explicit action-conditioned predictive-consistency objective is still needed.